A unified repertoire of indel mutational signatures in 6,975 tumors

Mo Liu1*#, Mi Ni Huang2*, Xue Ming Wu1*, Ying Yang1, Runtian Yao2, Qi Zheng1, Runxi Shen3, Steven G. Rozen4#

1 Sino-French Hoffmann Institute, School of Basic Medical Sciences, The First Affiliated Hospital, Guangzhou Medical University, Guangzhou, Guangdong 511436, China

2 Molecular Cancer Research Center, Zhongshan School of Medicine, Sun Yat-sen University Shenzhen Campus, Shenzhen, 518107, China

3 Imaging Platform, Broad Institute of Harvard and MIT, Cambridge, Massachusetts, United States

4 Department of Biostatistics and Bioinformatics, Duke University School of Medicine, Durham, North Carolina, 27710, United States

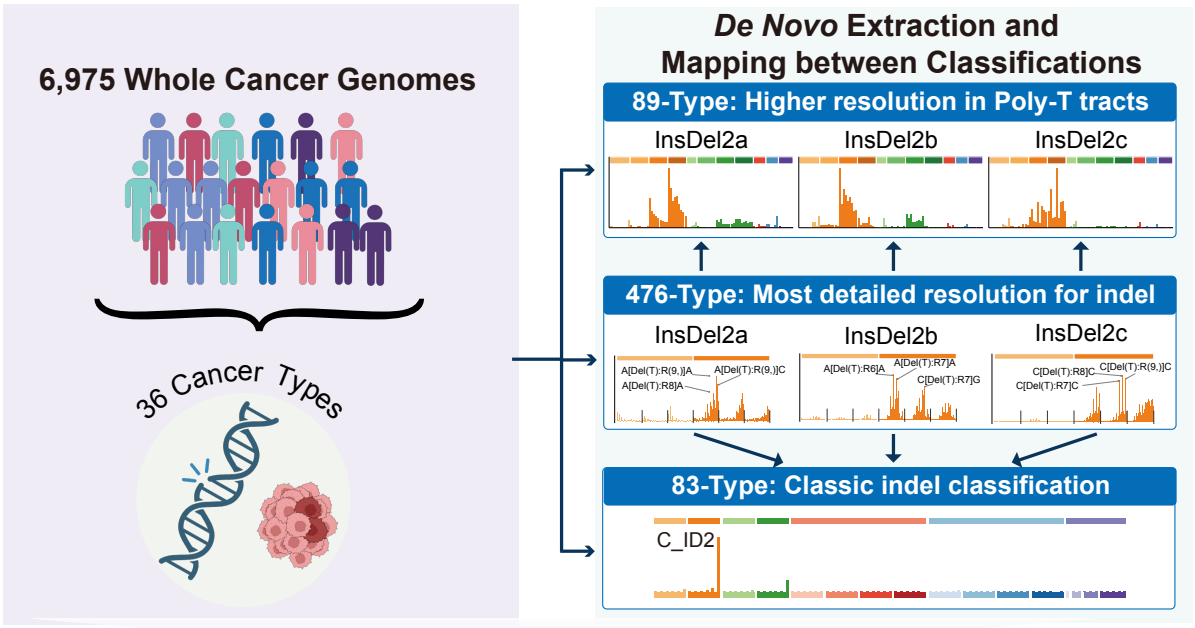
Corresponding authors: E-mails: mo.liu@gzhmu.edu.cn, steverozen@pm.me

* These authors contributed equally

Summary

We present a unified repertoire of indel mutational signatures in 6,975 tumors across 36 cancer types. Small insertions and deletions ("indels") are an important under-studied class of mutations. Koh, Nanda, and colleagues recently introduced two richer indel classifications, with 89 and 476 indel types. We identified signatures separately in the two new classification schemes and the widely used 83-type classification scheme using two algorithms. We identified 44 89-type signatures and their 476-type counterparts; 26 are new. Because lossless translation between the 83-type and newer schemes is impossible, we used tumors dominated by single signatures to exhaustively map correspondences across classifications. The 89- and, especially, the 476-type schemes distinguish signatures and spectra that the 83-type scheme cannot. We release software for generating and plotting spectra in the new classifications. An accompanying web site enables exploration of signatures and their cross-classification relationships and provides a service to construct spectra from VCF files.

Graphical Abstract



Highlights

1. Repertoire of indel signatures across 6,975 genomes in 3 indel classification schemes
2. Identified 26 novel 89-type and 11 novel 83-type classification indel signatures
3. Integrative topography across transcription, replication, and genic regions
4. Website and software for signature browsing and matrix generation from VCF files

Introduction

Somatic mutations are caused by various mutational processes and represent a driving force behind tumorigenesis and cancer development (Alexandrov et al. 2014). These mutations can result from either endogenous sources, such as 5-methylcytosine deamination or defective DNA repair mechanisms (Davies et al. 2017; Cooper et al. 2010; Grolleman et al. 2019; Boot et al. 2022), or exogenous sources, including exposure to chemical carcinogens in tobacco smoke or certain herbal medicines (Alexandrov et al. 2016; Ng et al. 2017; Dziubańska-Kusibab et al. 2020; Boot et al. 2020). Mutational signature analysis provides insights into cancer etiology, prognosis, prevention, and evolution. Mutational signatures can also serve as biomarkers for mutagenic exposures.

By "mutational signatures" we mean distinctive patterns of mutations left on genomes by mutagenic processes or exposures. They can be identified by exposing cultured cells, organoids, or experimental animals to suspected mutagens or perturbing DNA repair pathways and then sequencing the affected genomes (Boot et al. 2018; Huang et al. 2017; Kucab et al. 2019; Caipa Garcia et al. 2024; Riva et al. 2020). In addition, machine learning can identify mutational signatures as latent factors that explain the patterns of mutations in large collections of somatic mutation data (Alexandrov et al. 2020; Alexandrov et al. 2014; Nik-Zainal et al. 2012; Chen et al. 2024; Koh et al. 2025; Degasperi et al. 2022; Jin et al. 2024). For example, data mining of upper tract urothelial cancers (UTUC) from Taiwan initially identified the aristolochic acid (AA) single-base substitution (SBS) signature (Hoang et al. 2013; Poon et al. 2013). Subsequent analysis revealed that this signature was also present in bladder, kidney, and liver cancers (Ng et al. 2017; Poon et al. 2015). More recently, data mining of Chinese liver cancer genomes and experiments in cell culture showed that AA also generates double-base-substitution (DBS) signatures and, relevant to the current study, small insertion-and-deletion (indel) signatures (Figure 1) (Chen et al. 2024).

Mutational signature research has focused less on indel than on SBS signatures. Symptomatic of this, COSMIC v3.6 catalogs only 25 indel signatures compared to 78 SBS signatures. Nevertheless, indel signatures can yield important information for cancer classification, epidemiology, and treatment and for understanding DNA repair and mutagenic mechanisms. Examples include identification of tumors with homologous recombination deficiency or with mismatch repair deficiency (Huang et al. 2015; Koh et al. 2025; Davies et al. 2017) and a rare indel hypermutator phenotype due to a mutation in topoisomerase 2 alpha (Boot et al. 2022).

The first widely used classification of indels recognized 83 indel types (Alexandrov et al. 2020). Recently, Koh et al. (2025) proposed two classification schemes, one with 89 types of indel and one with 476 types (Figure 1). The 89-type scheme mainly grouped similar mutation types in the 476-type classification into coarser-grained types (Table S1). Koh et al. (2025) derived 37 indel signatures in the 89-type classification from 4,775 genomes across seven cancer types. This showed that the finer resolution of 1 bp T indels in the 89-type classificaton made it possible to distinguish clear different signatures that were indistiguishable in the older, 83-type system. However, challenges remain in connecting and translating between signatures in 83-type classification system and the new 89- and 476-type type schemes.

The older, 83-type scheme primarily categorizes indels based on the number of bases inserted or deleted, and for single-base insertions and deletions the identity of the deleted or inserted base (conventionally shown as pyrimidines, C or T). For these single-base indels, the classification also considers the number of flanking C or T residues (Figure 1A). Indels longer than 1 base are further classified by their occurrence within repetitive sequences (i.e. microsatellites, simple tandem

repeats). For deletions of ≥2 bases the 83-type system classification also encodes the presence of microhomology. (Microhomology is a hallmark of error-prone repair by non-homologous end-joining, notably in BRCA-deficient tumors.) For example, the 3-base deletion of TCA in (A**CA**|T**CA**|GG→A**CA**GG) exhibits a 2-base microhomology (CA).

A prominent distinction is that, unlike the 83-type classification scheme, both new classifications distinguish different types of indels of a single T or C according to the non-T (respectively, non-C) flanking bases. This substantially improves discrimination among mutational spectra and signatures containing single-base indels, which are present in most spectra and signatures. In more detail, the 83-type classification distinguishes deletions of single Ts only by the number of T's at the site of the deletion: 1, 2, 3, 4, 5, or ≥ 6 Ts (Figure 1A). By contrast, the 89-type classification distinguishes among deletions of T according to bases preceding or following the sequence of Ts at the site of the deletion. However, it groups together deletions of T from sequences of 1, 2, 3, or 4 T, and groups together deletions of sequences of 5, 6, 7, 8, or more Ts. Thus, for example, deletions of ATA→AA or ATTA→ATA are distinguished from deletions of CTA→CA or CTTA→CTA, but the deletions CTA→CA and CTTTTA→CTTTA have the same classification. The 476-type classification scheme provides fine-grained distinctions among deletions of a single C or T that combine the distinctions in the 83 and 89-type system. The 476-type classification scheme shows, for example, that in the signature in Figure 1 (InsDel23), the most prominent deletions are ATA→AA, CTA→CA, and GTA→GA. In the new classifications, the classifications of indels of 2 or more bases are substantially more complex than the single-base indels. Table S1 provides a mapping from the 476-type classification to the 89 and 83-type classifications, as well as multiple examples of each category of deletion.

Here, we analyzed somatic indels in 6,975 whole genomes to create a comprehensive collection of indel signatures in all three classification schemes using approaches based on hierarchical Dirichlet processes and on non-negative matrix factorization. We systematically matched the 476-type signatures to 89-type signatures by collapsing the 476-type signatures to 89-type signatures and comparing the collapsed signatures to signatures identified in the 89-type classification. We used tumor samples with spectra dominated by single signatures to map signatures in the 83-type classification to signatures in the other two classifications. We further profiled signatures' association with DNA replication timing, transcription and replication strand, and genic versus intergenic regions of the genome. We also assessed their associations with SBS signatures and with sex within individual cancer types. Finally, in addition to the supplementary information provided here, we provide a web site to facilitate exploration of these signatures (https://indelsig.net).

Results

Repertoire of indel signatures based on de novo signature discovery

We identified indel signatures in all three classifications from a total of 6,975 cancer genomes encompassing 36 cancer types using a method based on hierarchical Dirichlet processes (Liu et al. 2023), and a method based on non-negative matrix factorization (Islam et al. 2022). The extracted signatures were broadly consistent between the two methods (Table 1, Tables S2-S8). We identified 44 mutational signatures in the new 89-type classification and, independently, the 44 corresponding signatures in the 476-type classification. We assessed concordance between signatures identified in two classification systems by collapsing the 476-type signatures to their 89-type counterparts and

comparing them to signatures extracted in the 89-type classification. Signature discovery in the two classifications was highly concordant. Mean cosine similarity between corresponidng signatures in the two systems was > 0.989 (median > 0.994, minimum > 0.951). Further evidence of the signtures robustness derives from the identification of tumor spectra dominated by a single signature. For 35 of the 44 signatures, there were tumor spectra with cosine similarity > 0.95, and for 39 there were tumor spectra with similarity > 0.90 (Table 1).

Of the 44 89-type signatures, 18 were similar (cosine similarity ≥ 0.85, see Methods) to signatures described by Koh et al. (2025), and 26 were, to our knowledge, not previously described (Table 1). Of the 34 signatures we discovered in the 83-type classification, 20 were similar to COSMIC reference signatures, 3 were similar to signatures described by Jin et al. (2024), and 11 were not previously described (Table 1).

Because lossless translation between signatures in the 83-type and the two new classification schemes is impossible (Table S1), we used tumor mutational spectra dominated by individual signatures to systematically elucidate signature correspondences across classification schemes. Table 1 shows the correspondences, with details and discussion in https://indelsig.net/.

In several cases, the new indel classification systems were able to distinguish signatures that the older system merged. This is clearest for deletions of single T bases from tracts of ≥ 6 Ts. The tumors we analyzed had > 8.5 million of these deletions out of 18.4 million deletions in total. The COSMIC reference signature ID2 in the older, 83-type classification, consists exclusively of these deletions. The new indel classifications divide ID2 into three signatures. Among these, InsDel2b is dominated by deletions of T from tracts of 5 to 7 Ts. InsDel2a and InsDel2c are both dominated by deletions of single Ts from stretches of ≥ 8 Ts. However, they are distinguished by InsDel2a's preference for flanking A bases and InsDel2c's preference for flanking C bases. Mutations attributed to InsDel2b and InsDel2c are strongly enriched among MSI-H tumors. Furthermore, the counts of mutations attributed to these signatures are correlated with the counts of mutations attributed to the MSI-H-associated SBS signatures SBS6, SBS26, and SBS44 (Figure 4, Figure S1).

The new indel classification systems were also able to distinguish insertion signatures that the older system merged. The tumors we analayzed had 3.3 million insertions of single T bases into tracts of ≥ 5 Ts. In the 83-type system this constitutes a single mutation class, and the COSMIC reference signature ID1 consists exclusively of this class of insertions. The new indel classifications divide ID1 into three signatures. Among these, InsDel1a has the largest proportion of insertions into tracts of ≥ 8 Ts and InsDel1b has the smallest proportion, while InsDel1c has an intermediate proportion. Compared to InsDel1b, InsDel1a and InsDel1c have larger proportions of A to the left of the poly-T tract and G to the right, and smaller proportions of C to the left of the poly-T tract with C or A to the right. InsDel1b is further distinguished from InsDel1a and 1c in that several tumors with high proportions of InsDel1b (> 0.4) have mutations in the proofreading domain of polymerase epsilon (Table S9).

Two cases where the 89-type classification misses clear differences

While the 89-type classification system on the whole better discriminates signatures, in two cases the 83-type and 476-type systems clearly differentiated signatures that were almost identical in the 89-type system. One case consisted of InsDel13 and InsDel_C. These are nearly identical in the 89-type system, which groups deletions of single Ts in stretches of 1, 2, 3, or 4 Ts as a single deletion type. However, the 83- and 476-type systems show that InsDel_C has deletions of 1 T from ATA (ATA→AA), while InsDel13 has deletions of T from ATTA (ATTA→ATA, Figure S2A). The other case consisted of InsDel11 and InsDel16. InsDel11 has insertions of one C next to an isolated

C (C with non-Cs on each side, e.g. ACG→ACCG) and insertion of one T next to an isolated T (e.g. GTG→GTTG). InsDel16 has insertion of one C between non-Cs (e.g. AT→ACT), and insertion of one T between non-Ts (e.g. AG→ATG, Figure S2B). Again, the 89-type system does not capture these differences, while the 83 and 476-type systems do.

Clarified signatures of aristolochic-acid-induced deletions

InsDel23 is caused by aristolochic acid (AA) exposure and the 476-type signature shows that most AA deletions involve 5'TA3'/3'AT5' → A/T (Figure 1, and see https://indelsig.net). This pattern is not visible in either the 83-type or 89-type classification. One might speculate that this pattern is related to the adenine adducts caused by AA, which are also thought to be responsible for the predominance of A→T single base substitutions in the AA SBS signature and for the frequent involvement of adenines in the AA DBS signature.

APOBEC3A activity and InsDel_Aβ

We identified InsDel_Aβ, an indel signature that in the 476-type classification consists primarily of the 1-bp C deletions TCA→TA and TCT→TT (Figure S2C, right sub-panel). Thus, InsDel_Aβ targets the same sequences as the canonical APOBEC SBS signatures SBS2 (TCA→TTA and TCT→TTT) and SBS13 (TCA→TGA and TCT→TGT). Experiments in cell culture showed that, in addition to single nucleotide substitutions, APOBEC3A induces indels, mainly TC→T (DeWeerd et al. 2022). In parallel, we discovered the corresponding 83-type indel signature, ID_A, which is more similar to the experimental spectrum than C_ID9. Notably, C_ID9/InsDel9 in all three classification systems has many deletions of single thymines, which are absent in ID_A/InsDel_Aβ (Figure S2C). In addition, the InsDel9 476-type signature has many deletions of a single C following A or G, while InsDel_Aβ has almost exclusively deletions of a single C following T. This signature was also noted in Koh et al. (2025), who denoted it InD9a. We propose that this signature dominated by TCA→TA and TCT→TT is distinct enough from InsDel9to warrant the new identifiers ID_A/InsDel_Aβ.

MSS tumors: single-thymine indels in poly-T tracts

De novo signature discovery using both HDP and NMF in all three signature classification systems identified signatures with similar proportions of deletions and insertions of a single thymine in poly-T tracts. These occur almost exclusively in MSS tumors. We termed these ID_N (the 83-type signature) plus InsDel_Nα and InsDel_Nβ (the 89- and 476-type signatures). Previous analysis in the 83-type classification allocated insertions and deletions of single thymines in poly-T tracts to separate signatures ID1 and ID2, and did not differentiate between signature occurrence in MSS versus MSI-H (Alexandrov et al. 2020). We first focus on InsDel_Nβ which occurs in 2,629 of 6,770 MSS tumors (39%) in diverse cancer types and in only 9 of 175 MSI-H tumors (5%, Table S10). Roughly half (45%) of the MSS tumors with InsDel_Nβ also have InsDel1a, and InsDel1a also occurs without InsDel_Nβ in 2,596 MSS tumors (38%) and 35 MSI-H tumors (20%). Thus, many MSS tumors have roughly equal proportions of thymine deletions and insertion in long poly-T tracts (InsDel_Nβ), but other tumors have varying proportions of additional single-T insertions (InsDel1a), and some tumor spectra are completely dominated by these single-T insertions. The other signature with approximately equal proportions of insertions and deletions of a thymine in a poly-T tract, InsDel_Nα, is rare in both MSS and MSI-H tumors, occurring in only 55 of 6,770 MSS tumors and 2 of 175 MSI-H tumors.

Indels in poly-T tracts and microsatellite mutations in MSI-H tumors

MSI-H tumors were enriched for InsDel2a, 2b, and 2c, which represent single-thymine deletions from poly-T tracts, and were also notably enriched for InsDel7, InsDel_D, and InsDel_J (Figure 4A).

InsDel7, like InsDel2a/b/c, is dominated by single-thymine deletions from poly-T tracts, but with a distinct distribution of sequences flanking the poly-T tract. InsDel_J consists almost entirely of deletions of single dinucleotides from dinucleotide microsatellites, e.g. deletion of CA from $(CA)^5$, or deletion of GT from $(GT)^8$ [see also entries for the 476-indel-type category Del2:U2:R(5,) in Table S1]. Unlike the other signatures enriched in MSI-H tumors, InsDel_D consists mainly of insertions. Single-thymine insertions into poly-T tracts make up 71% of the signature, mostly into tracts of ≥ 8 Ts. InsDel_D is distinguished from InsDel1a/b/c by a larger proportion of insertions of repeat units into microsatellites, mostly single dinucleotide insertions into dinucleotide microsatellites, e.g. insertion of AC into $(AC)^5$, or insertion of CA into $(CA)^8$, in a tumor with cosine similarity 0.9997 to InsDel_D in the 476-type classification (SP102133).

The new indel classifications reveal a more specific signature of RNase H2 deficiency

Cells with RNase H2 deficiency cannot efficiently remove ribonucleotides that had previously been incorporated into genomic DNA, and this sets the stage for topoisomerase-1 (TOP1) associated mutagenesis. This was previously thought to generate indel signature ID4 (Reijns et al. 2022; Koh et al. 2025; Jin et al. 2024). However, we detected a new indel signature, InsDel_F that corresponds better to experimental RNase H2 deficiency data than ID4/InsDel4 (Figures 6, S5). The profile of deletions of ≥2 bases is very different between InsDel_F and InsDel4: InsDel_F has distinctively high proportions of (i) 2-base deletions with microhomology (almost all TCT → T or TGT → T) and (ii) 2-base deletions from paired 2-base doublets (most often CTCT → CT). InsDel4 has much lower proportions of these two deletion types and also has additional prominent deletion types that are essentially absent in InsDel_F. These additional deletion types are mainly deletions of 3 or 4 bases with microhomology and deletions of 3 or 4 bases from paired doublets (e.g. TCTTCT → TCT, GTTTGTTT → GTTT). The spectra in RNase H2 deficient cell lines consist of signatures due to mutational processes intrinsic to the parental cell line combined with high proportions of mutations characteristic of InsDel_F (Figure S5). Like InsDel_F, these cell-line spectra have 2-base deletions with microhomology and 2-base deletions from paired 2-base doublets. These spectra conspicuously lack the 3- and 4-base deletions present in InsDel4. Similarly, RNase H2 deficient mouse tumors have high proportions of the 2-base deletions characteristic of InsDel_F and lack the 3 and 4 base deletions present in InsDel4. Thus, ID_F/InsDel_F is clearly distinct from InsDel4 (cosine similairy XXX, XXX and XXXX in the 83- 89- and 476-type classifications, respectively) and it was extracted by both the HDP and NMF-based methods in all 3 indel classification schemes. All available evidence indicates that it more accuratly reflects the mutational consequences of RNase H2 deficiency than InsDel4.

Associations between indel signatures and SBS signatures

We further analyzed the correlations between the indel signature activities we derived and the SBS signature activities from Degasperi et al. (2022) (Table S11). We first computed a matrix of Spearman correlations between the numbers of mutations attributed to each SBS signature and to each indel signature over all SBS-indel signature pairs. Correlations that were not significant (Benjamini-Hochberg FDR ≥ 0.05) were set to 0. Agglomerative hierarchical clustering (Ward's method) on the rows and columns of this matrix, using as distance 1 – the Pearson correlation between the rows (or columns) of the Spearman correlation matrix, identified several clusters (Figure 4B). The APOBEC-related cluster includes SBS2 and SBS13 and showed moderate correlations with ID_A

and InsDel_Aβ. The homologous recombination deficiency[(HRD) cluster, consisting of SBS3 and SBS8, was strongly associated with InsDel6 and InsDel9 (Figure 4B, "HR defects"). UV-related signatures, such as SBS7a, SBS38, InsDel13, and InsDel_H, also clustered together. The cluster related to mutations in the proofreading domain of the polymerase epsilon (*POLE*) gene linked SBS10a with InsDel1b and InsDel_K. Additional analysis of tumor somatic mutations confirmed that these signatures are associated with dysfunctional *POLE* proofreading plus, in some tumors, mismatch repair deficiency (Table S9).

Within the mismatch repair deficiency (MMR) cluster, we observed a strong network of correlations between SBS6, SBS26, SBS44, and a group of indel signatures, including C_ID2, InsDel2b, InsDel2c, InsDel7, InsDel_D, and InsDel_J (Figure 4, "MMR defects"). Another notable group included InsDel_B, InsDel5a, and InsDel5b, which co-occurred with liver-cancer associated SBS signatures SBS12 and SBS16.

The 89-type taxonomy also allowed us to better resolve the relationships within previously broader categories. For instance, the three signatures derived from C_ID1 (InsDel1a, InsDel1b, and InsDel1c) each showed distinct patterns of correlation to SBS signatures (Figure 4B). InsDel1a and the 83-type signature C_ID1 had similar correlation profiles, with both having modest correlations (Spearman ρ 0.1–0.2) to SBS1, SBS18, SBS2, SBS35, and SBS13. Among these, SBS1, SBS18, and SBS35 are in the gastrointestinal (GI) cluster. In contrast, InsDel1b was correlated only with SBS10a ($\rho = 0.30$), which is caused by *POLE* proofreading errors, and InsDel1c was weakly correlated with SBS12 and SBS16 ($\rho \approx 0.12$), which are common in liver cancer. The correlation of C_ID1 and InsDel1a with SBS18, which was absent for InsDel1b and InsDel1c, is consistent with a possible contribution of reactive oxygen species to InsDel1a (Figure 4B).

These correlation patterns provide insight into the possible origins of previously uncharacterized SBS signatures. For example, SBS92 closely mirrored the correlation profile of SBS4, with strong associations with C_ID3 and InsDel3b, further supporting a link with tobacco smoking (Figure 4B, "Tobacco Smoking") (Lawson et al. 2020).

In the GI cluster, which includes signatures common in gastrointestinal tumors, signatures C_ID18/InsDel18, ID_N/InsDel_Nβ were associated with SBS signatures typically seen in the GI tract. The GI cluster also contained the signatures C_ID14/InsDel14. These were strongly correlated with SBS93, which has been found almost exclusively in stomach and esophageal tumors, and which has transcriptional strand bias suggesting thymine damage (Islam et al. 2022; Moody et al. 2021).

Topography of indel mutational signatures <STEVE REVIEW>

We evaluated the interplay between the 83-type signatures we identified and several genomic topographical features. Transcription-coupled nucleotide excision repair causes transcription strand asymmetries, since this pathway preferentially removes bulky DNA adducts on the transcribed strand of active genes (Mellon, Spivak, and Hanawalt 1987). There was evidence of transcriptional strand asymmetry in 17 out of the 21 signatures with sufficient data to assess (Figure 5A, Figure S3A, Table S12). Signatures attributed to exogenous mutational processes, such as C_ID3 (tobacco smoking exposure) and C_ID18 (colibactin exposure), consistently showed transcription strand bias that could be explained by transcription-coupled repair of bulky adducts on guanines (tobacco) or adenines (colibactin), which accords with the likely mechanisms of the correspoing SBS mutational signatures. C_ID14, which is of unknown etiology but strongly correlated with SBS93

(see *Associations between indel signatures and SBS signatures*), also showed transcription strand bias, suggesting that it may also arise from DNA damage subject to transcription-coupled repair.

Note on C_ID14, 2026-10-06

Note about C_ID14, probably incorporate into text in simplified form: SBS93 is mostly T>C, T>G, and T>A (69% of the COSMIC v3.4 profile, https://cancer.sanger.ac.uk/signatures/sbs/sbs93/). Islam et al. 2022 (https://doi.org/10.1016/j.xgen.2022.100179) describe it as "characterized by T>C and T>G mutations with a strand asymmetry consistent with damage on pyrimidines for TpTpA contexts". So the published interpretation is thymine damage, not purine damage. C_ID14 itself is 55% INS:C:1:0 (insertion of a single C outside a C tract, Table S2), and the pipeline labels it Transcribed, meaning the inserted C is on the transcribed strand. An inserted base is not the damaged base, so C_ID14's strand bias does not by itself tell us which base was adducted. That is why the sentence above does not name one.

Note about flanking bases of the C insertions in C_ID14: Orientation: in mSigSpectra (categorize_1_justified_indel.R, lines 181-185), a 1-base indel of A or G is reverse complemented and its flanks are swapped, so 476-type contexts are always written 5′ to 3′ on the strand carrying the C or T. Signature: in InsDel14, the 476-type counterpart of C_ID14 (Table S6), single-C insertions outside a C tract make up 42% of the signature. Within those, A[Ins(C)]A accounts for 61%, A[Ins(C)]T for 31%, and T[Ins(C)]T for 6%. Tumor: CPCT02340014T has all of its indel exposure assigned to C_ID14 in Table S11. Its annotated indel VCF (fmh-unfiltered_vcfs/CPCT02340014T.annotated.indel.vcf.gz, examined with mSigSpectra::build_one_file_rosetta() and its internal .revcomp_single_tc_long_visual()) has 4,029 indels, of which 3,425 are PASS. All 1,600 Ins(C):R0 indels are PASS, so they are 47% of the PASS indels. **On the C strand, 91% of these C's are preceded by a run of at least 4 A's (12 C's follow a run of exactly 3 A's, 331 a run of 4, and 534 a run of 5), and 85% are followed by 0 to 2 A's.** So the dominant event is insertion of a C at or near the 3′ end of a poly-A tract of 4 or more. On the other strand, this is insertion of a G at or near the 5′ end of a poly-T tract. A[Ins(C)]A and A[Ins(C)]T are the same kind of event. They differ only in whether the C lands just before or just after the last A.

The topography pipeline labels C_ID14 Transcribed. This puts the inserted C, and the poly-A tract next to it, on the transcribed strand, and the poly-T tract on the untranscribed strand. There are two possible explanations. One is that the C insertions arise from lesions on the poly-T strand. The other is that they arise from polymerase misincorporation or slippage at the boundaries of poly-T tracts, without DNA damage.

The transcription strand asymmetry itself favors the first explanation. Transcription-coupled nucleotide excision repair removes lesions from the transcribed strand. Lesions that persist, and the mutations they cause, are therefore enriched on the untranscribed strand, which is where the poly-T tract is. We are unaware of any reason that polymerase misincorporation or slippage events at the boundaries of poly-T tracts would differ in frequency between the transcribed and untranscribed strands. We note that C_ID14 has the strongest transcription strand asymmetry in Figure 5A, with odds ratios from 1.5 to 1.75.

The correlation of C_ID14 with SBS93 is consistent with this. Islam et al. 2022 (https://doi.org/10.1016/j.xgen.2022.100179) describe SBS93 as having "a strand asymmetry consistent with damage on pyrimidines for TpTpA contexts". This points to damage on thymines, although they do not identify the lesion or say that it is a bulky adduct. Of course, the correlation does not by itself show that C_ID14 and SBS93 have the same cause.

If C_ID14 and SBS93 share a lesion, the C insertions occur at the opposite end of the poly-T tract from the TpTpA position that Islam et al. associate with SBS93 damage (see the backup material below).
Backup material, possibly for a supplementary note: where an SBS93-type lesion would be relative to the C_ID14 insertion. Example with a run of 5 and the C inserted just before the last A:
steve note: this diagram seem wrong; the position of the C insertion should be one to the left; we need to dig deeper. Is there an A to the left on the bottom strand? how strong is the TTA context for the SBS

```
                left end                          right end
                   |                                  |
 top     5'-  X  [ A  A  A  A  A ] C  A  ...  -3'   C strand: on the TRANSCRIBED strand
                   :  :  :  :  :   :  :
 bottom  3'-  X' [ T  T  T  T  T ] G  T  ...  -5'   poly-
T strand: on the UNTRANSCRIBED strand
                   ^                 ^
                   |                 |
     (1) SBS93-type damaged T        (2) where C_ID14 adds the extra base
         (TpTpA context)                 (C on top, G on bottom)
```

The bottom strand runs 5′ to 3′ from right to left. Read in that direction it is ... T G T T T T T X′ ...

```
AAAA[Ins(C)]A

AAAA[Ins(C)]T
```

(1) Where the SBS93 lesion would be. Islam et al. place the SBS93 damage on the middle T of a TpTpA, which is a T with another T on its 5′ side and an A on its 3′ side. On the bottom strand, 3′ is to the left. So the only T in the run that can have an A on its 3′ side is the leftmost one, and only if X′ is A. That puts the SBS93-type damaged T at the left end of the run.
(2) Where C_ID14 adds a base. The extra C (G on the bottom strand) is at the right end of the run.
If the poly-T strand is the template, the new top strand is made from left to right. The polymerase would pass the SBS93-type position (1) first, copy the rest of the run, and add the extra C at (2), about 5 bases later.
This matters only if C_ID14 and SBS93 have the same lesion. SBS93's TpTpA context is what places that lesion at the left end. If the lesion behind C_ID14 is a damaged T at or next to position (2), there is no mismatch. Nothing about the strand asymmetry requires the lesion to be at a TpTpA. To check: the bases flanking the poly-T tract in the CPCT02340014T VCF.

Another exogenous mutational signature C_ID13, which is attributed to UV exposure, need cite – is this associatioin with sbs7 in melanoma? does this have have the same transcribed strand polarity as CID_14? consistently showed transcription strand bias that could be explained by transcription-coupled repair of damaged cytosines, which accords with the mechanisms of the correspoing SBS mutational signatures (Vrieling et al. 1989).

In 83-type signatures associated with defective endogenous mutational processes, enriched mutations on the untranscribed strand were observed for C_ID1 (slippage during DNA replication) this contradicts a statment about id14 no slipage difference between untranscribed / transcribed strands and C_ID5 (clock-like signature). Enrichment for mutations on the transcribed strand were observed for ID_F (topoisomerase-1 [TOP1] associated mutagenesis stemming from RNase H2 deficiency). In addition, two signatures associated with defective MMR, C_ID7 and ID_K, had transcription strand asymmetries. what are these, and what direction are these?

Needs review: The statements in the paragraph above that mutations are "enriched on the transcribed strand" or "on the untranscribed strand" (C_ID13, C_ID1, C_ID5, ID_F) use the pyrimidine-strand shorthand of Otlu et al. 2023 (https://doi.org/10.1016/j.celrep.2023.112930) without saying so. The C_ID3/C_ID18 sentence has been rewritten in terms of adducted purines, and the C_ID14 sentence no longer states a direction. Also, "This is because UV damage occurs on cytosine rather than guanine" is not correct as written (UV damages adjacent pyrimidines, C or T) and garbles Otlu et al.'s "consistent with damage from ultraviolet light on cytosine". See the note in the Figure 5 legend.

Note: What the strand labels mean in terms of bases for each signature. In ID83 labels, the last number is the length of the homopolymer before the indel minus 1 for deletions, and the length of the homopolymer for insertions (mSigSpectra gen_83_type_string.R). Strand labels come from build_fig5/trans.strand.cancer_proportion.xlsx, and "on strand X" means the C or T bases of the indel are on strand X. C_ID13 (UV, Untranscribed): 46% single-T deletions, mostly DEL:T:1:1 (29%), deletion of one T from a TT. So the TpT, the classic UV dimer site, is on the untranscribed strand. Here the counted pyrimidines are the damaged bases, so this matches damage persisting on the untranscribed strand after transcription-coupled repair. C_ID1 (replication slippage, Untranscribed): 60% single-T insertions, mostly INS:T:1:5+ (56%), a T added to a poly-T tract of 5 or more. So the poly-T tract that gained a T is on the untranscribed strand, and the poly-A tract that gained an A is on the transcribed strand. No DNA damage is implied. C_ID5 (clock-like, Untranscribed): spread out. Single-T deletions are 36% (DEL:T:1:0 to 1:3, i.e. from T runs of 1 to 4), single-C deletions 15%, microhomology deletions 15%, single-T insertions 12%. The microhomology and longer deletions are only counted if they contain only pyrimidines or only purines. ID_F (TOP1/RNase H2, Transcribed): mostly 2-bp deletions, DEL:MH:2:1 (25%) and DEL:repeats:2:0 to 2:2 (32%). A 2-bp deletion is counted only if both bases are pyrimidines or both are purines, so mixed deletions such as TG or CA are excluded from the strand test. We do not yet know what fraction of ID_F indels were counted. C_ID7 (defective MMR, Transcribed): 42% single-C deletions, mostly from C tracts of 6 or more (DEL:C:1:5+, 24%) or 5 (DEL:C:1:4, 10%), plus single-T deletions from T tracts of 5 (DEL:T:1:4, 10%). So the poly-C or poly-T tract that lost a base is on the transcribed strand, and the poly-G or poly-A tract on the untranscribed strand. ID_K (POLE, Transcribed): 80% single-T deletions, mostly from T tracts of 5 (DEL:T:1:4, 48%) or 4 (DEL:T:1:3, 24%). So the poly-T tract that lost a T is on the transcribed strand, and the poly-A tract on the untranscribed strand. For C_ID1, C_ID7, and ID_K, the result is a statement about which strand carries the poly-T (or poly-C) tract, not about damage. Signature compositions are from Table S2.

[doublecheck that the above is neutral for skew of polyt vs polya on transcribed /non-transcribed]

There was evidence of replication strand asymmetry in 12 out of the 26 signatures with sufficient data to assess (Figure 5B, Figure S3B, Table S12). Replication strand asymmetry was less common than transcription strand asymmetry, and none of the tobacco-, UV-, or colibactin-associated signatures (C_ID3, C_ID13, C_ID18) showed it.

[do the corresponidng sbs have replication strand asymmetry?]

ID_K, which is enriched in tumors with POLE proofreading defects, was consistently enriched for leading strand mutations. This is consistent with SBS signatures SBS10a and SBS10b, which are also driven by POLE proofreading defects, thus further supporting a link between POLE proofreading defects and leading-strand mutagenesis. sentence not clear do sbs10a/b have repication strand asymmetry; same issue about what the asymmetry means for the indels; is this 1bp T deletions from shorter stre

Note: Figure 5 (main_figures/Figure5.pdf) is now a copy of build_fig5/Figure5.pdf in the code repo. Its panels A to C use the topography pipeline's labels (build_fig5/assign_asymmetry_type.R), which also underlie Figure S3, including ID_K on the leading strand. This changed two signatures relative to the earlier hand-edited Figure 5. In panel B, ID_L is now Lagging (its only cancer type with $P < 0.05$ is Pancreas, lagging). In panel C, ID_B is now Mix (8 cancer types with $P < 0.05$ are intergenic and 1, Breast, is genic). Mini confirmed by email on 2026-10-07 that repli.strand = -1 is the lagging strand and that the pipeline's leading/lagging orientation (as in Figure S3B) is correct.

We also examined the mutation enrichment of 83-type signatures in genic and intergenic regions (Figure S3C). Most signatures showed mutation bias towards intergenic regions, while 6 signatures showed enrichment in genic regions (Figure 5C). Among the 6 signatures, three were associated with defective endogenous mutational processes, including C_ID8 (repair of DNA double strand breaks by NHEJ), C_ID17 (TOP2A K743N mediated mutagenesis) and ID_F (TOP1-mediated mutagenesis). Another 3 signatures with mutations enriched in genic regions were of unknown etiology (C_ID4, ID_E and ID_L).

The effect of DNA replication timing was also observed in many 83-type signatures (Figure 5D, Figure S4, Table S13). Most signatures were consistently enriched in late-replicating regions across different cancer types, while only two signatures, C_ID10 and C_ID17 (TOP2A K743N mediated mutagenesis), showed enrichment in early-replicating regions. Five signatures were unaffected by replication timing in the majority of cancers presenting them: C_ID1 (Slippage during DNA replication), C_ID5 (Clock-like signature), C_ID18 (Colibactin exposure), and two defective MMR signatures (C_ID7 and ID_K). Interestingly, several signatures showed cancer-specific enrichment in replication timing, including C_ID8 (Repair of DNA double strand breaks by NHEJ) and ID_F (TOP1-mediated mutagenesis). For example, ID_F showed enriched mutations in late replication regions in bladder cancer, while in skin cancer it showed the opposite trend. For 89-type signatures, we conducted similar analysis without simulation data. The majority of signatures showed enrichment in late-replicating regions, which is consistent with what we observed for 83-type signatures (Table S14). InsDel_L, the corresponding signature of ID_L, showed the same stomach cancer specific enrichment in early replication regions, which strengthens the correspondence between 83-type and 89-type signatures.

Do we really need Table S14? It gives P values for the regression of indel counts on replication-timing decile, but there is no simulation. So, unlike Table S13, it does not test whether a trend differs from what the background mutation process would produce.

Demographic Associations with Sex and Age STEVE REVIEW

To explore potential sex biases in mutational processes, we systematically compared signature activities between males and females using two-sided Fisher's exact tests. After applying Benjamini–Hochberg adjustment ($q < 0.2$), nine signature-cancer type associations with sex were identified (Figure S6A, Table S15). <parallel structure>Male-enriched mutational processes predominated in lung, liver, and "Other" cancer categories. Specifically, lung cancer showed significant male association with InsDel3b (OR = 0.367, q = 0.061) and C_ID3 (OR = 0.397, q = 0.188), signatures previously linked to tobacco exposure, consistent with higher smoking prevalence among men. In liver cancer, a similar male predominance was observed for InsDel5a (OR = 0.481, q = 0.118; clock-like) and InsDel8/C_ID8 (OR = 0.271, q = 0.184; associated with DSB repair by NHEJ and age). <mentioned anywhere else?>~~The "Other" tumor group exhibited male bias in C_ID19 (OR = 0.424, q = 0.188) and InsDel19b (OR = 0.448, q = 0.197).~~ In contrast, female-enriched signatures were found in esophageal and lymphoid malignancies, most notably in esophageal cancer for InsDel1a (OR = 3.85, q = 0.061) and C_ID1 (OR = 3.22, q = 0.188), as well as InsDel1c (OR = 2.63, q = 0.194) in lymphoid cancers. No additional signature–cancer type combinations reached a q-value below 0.2. Collectively, these findings suggest that specific mutational processes and their underlying biological processes might be modulated by sex, shaping distinct patterns of cancer susceptibility and progression.

> **Reviewer note**
>
> Problem: Nothing on susceptibility or progression was measured. Proposed: End the sentence at "modulated by sex".

> **Comment 0, Mo Liu, 2026-08-21**
>
> The clock-like association was described in the following paragraph. We can either not mention the clock-like in this part, or bring the age-association part earlier.

To characterize the "clock-like" behavior of these signatures, we examined correlations between signature activity and patient age at diagnosis using Spearman correlation analyses (Figure S6B). We found that C_ID1, InsDel1a, C_ID5, and InsDel5a showed robust, age-associated "clock-like" mutagenesis in multiple cancer types, as well as across the overall genome. More specifically, C_ID5 and InsDel5a correlated with age in prostate, pancreas, liver, and kidney cancers, while C_ID1 and InsDel1a were age-associated in breast, prostate, and pancreas cancers. Other clock-like signatures demonstrated pronounced tissue specificity. For example, InsDel9 showed a more attenuated relationship between age and mutational accumulation in prostate cancer, while InsDel10 displayed clear clock-like properties in pancreatic cancers. Additionally, some signatures, such as InsDel5b, InsDel5c, and C_ID4/InsDel4, were correlated with age when considering all tumor types collectively but did not show significant associations in any single cancer type (Table S16). However, these broad pan-cancer associations likely reflect weak or heterogeneous age dependencies spread across multiple tumor types, rather than strong, tissue-specific age effects.

> **Reviewer note**
>
> Problem: "Robust" needs the correlation strengths or a q threshold in the text. Proposed: Give rho and q for these four signatures, or write "showed significant age association (q < X)".

Taken together, these analyses reveal a demographic landscape of mutational mechanisms, indicating that genomic instability is sculpted by both sex-specific biological influences and tissue-dependent aging processes.

> **Reviewer note**
>
> Problem: Exposure differences could explain the sex associations entirely, and "sculpted" and "landscape" are rhetorical. Proposed: "Together, these analyses identify sex- and age-associated indel signatures within individual cancer types."

Discussion

In this study, we analyzed somatic indels in 6,975 whole-genome tumor samples spanning 36 cancer types and established a comprehensive repertoire of indel mutational signatures across the 83-type, 89-type, and 476-type classification systems. By combining de novo signature discovery with systematic cross-classification mapping, we show that the 89-type and 476-type taxonomies provide substantially greater resolution than the conventional 83-type scheme, particularly for single-base insertions and deletions in context-dependent sequence environments. This expanded framework not only clarifies correspondences among previously reported signatures, but also enables the identification of novel signatures, including ID_F and InsDel_F, which are associated with TOP1-related mutagenesis and RNase H2 deficiency. In addition, we identified six 89-type and 476-type signatures, together with four 83-type signatures, that are strongly enriched in high-mutation-burden tumors, and further refined the interpretation of indel processes linked to mismatch repair deficiency and/or POLE/POLD proofreading dysfunction. By analyzing a larger and more diverse cancer genome cohort, we not only expanded the existing indel signature database, but also systematically linked and unified the nomenclature of the 83-type and 89-type signatures.

> **Reviewer note**
>
> Problem: This result is not reported in the Results section. Proposed: Add the result to Results (with a figure or table reference) or drop the sentence.

A major strength of this study is the systematic mapping between the conventional 83-type scheme and the newer 89-type and 476-type indel taxonomies. Because translation across these classification systems is inherently imperfect, direct comparison of signatures is not straightforward. By leveraging tumors dominated by single mutational processes, we established robust correspondences among signatures identified in all three frameworks. This unified mapping not only improves continuity with legacy studies based on the 83-type classification, but also provides a practical reference structure that will facilitate future indel signature analyses using the more informative 89-type and 476-type schemes.

> **Reviewer note**
>
> Problem: Table 1 has 9 of 44 signatures without a tumor at > 0.90 similarity and several matches required zeroing poly-T insertions. Proposed: "we established correspondences for most signatures across the three frameworks".

The new indel framework also refined the interpretation of several biologically important mutational processes. Notably, the new TOP1-associated signatures, ID_F and InsDel_F, were more consistent with RNase H2 deficiency than previously recognized indel patterns. The enrichment of microhomology-mediated deletions and paired doublet deletions in these signatures supports a mechanistic link between RNase H2 dysfunction and a characteristic indel spectrum. Similarly, we identified multiple signatures strongly associated with microsatellite instability, and the finer resolution of the new taxonomies allowed us to distinguish MSI-related signatures that were previously aggregated or less clearly resolved. Given the clinical relevance of MSI and mismatch repair deficiency, this improved annotation may have direct utility for tumor classification and therapeutic stratification.

> **Reviewer note**
>
> Problem: Nothing in the paper tests utility for classification or stratification. Proposed: "this improved annotation may be useful for tumor classification".

Our integrative analyses also examined the relationships between indel signatures and SBS signatures. The clustering of APOBEC-, UV-, HR-, MMR-, and POLE-related signatures is consistent with shared or linked mutational processes across mutation classes, supporting the idea that indel mutational signatures often arise in the context of broader mutational programs. These associations provide an additional layer of biological interpretation and help connect indel signatures to established mutagenic pathways.

We also investigated the genome-wide topographical features associated with indel signatures, including transcriptional strand bias, replication strand bias, genic distribution, and replication timing. Notably, because the COSMIC catalog does not provide such topographical annotations for many indel signatures, our analysis in this large cohort helps fill an important gap in the current knowledge base. The observed associations highlight the influence of DNA repair and replication dynamics on indel formation and suggest that genomic context contributes to the emergence and distribution of distinct indel patterns. Together, these analyses provide a more comprehensive view of the biological determinants of indel mutagenesis.

> **Reviewer note**
>
> Problem: Vague, and the comment thread below questions it. Mo notes that COSMIC fully annotates only ID6 with all four features. Proposed: "COSMIC provides all four topographical annotations for only one indel signature (ID6), so our analysis substantially extends the available annotations."

Comment 1, Unknown Author, 2026-07-02

Reply to Mo Liu (05/29/2026, 16:03): "…" Hi Steve, I summarized the topographical annotations available on the COSMIC website. In our paper, we examined the following four features: replication strand asymmetry, transcription strand asymmetry, genic strand asymmetry, and replication timing. On COSMIC, except for ID6, no signature was fully annotated with all four features. So our work indeed provides important topographical annotations for the 83-type signatures. I've summarized the information in the attached file. Please let me know if you need more information. Thanks,Mo See COSMICv3.6 ID topographical annotation.xlsx

Comment 2, Mo Liu, 2026-05-29

I update via email.

Comment 3, Steve Rozen, Ph.D., 2026-05-28

Which ones do we supply that COSMIC does not, and are there any discordances between what we get and COSMIC for signatures that COSMIC does have topographical annotations for?

Comment 4, Mo Liu, 2026-05-25

This is true. But not sure if this worths mentioning?

A potential source of technical variation in our analysis is the use of data generated by different variant-calling pipelines used for the PCAWG (Pan-Cancer Analysis of Whole Genomes) and Hartwig cohorts. We observed a higher frequency of indels occurring in sequence contexts containing more than nine repeat units in the Hartwig data, raising the possibility that differences in variant calling, local realignment, or indel representation contributed to this discrepancy. To assess and minimize the potential impact of this bias, we performed two additional analyses. First, we excluded indels associated with more than nine repeat units to improve comparability with the PCAWG calling pipeline. Second, for indels occurring in repetitive sequence contexts, we normalized their representation by moving each indel to the leftmost possible position within the sequence context. Even so, we cannot exclude residual caller- or cohort-specific effects, particularly for indels in highly repetitive sequence contexts and for signatures supported by relatively few mutations. Nevertheless, the overall consistency of the dominant signatures across datasets, together with the robustness of the results after filtering and representation normalization, suggests that variant-calling bias is unlikely to substantially affect the main conclusions of this study.

Reviewer note

Problem: Needs a number or a supplementary reference. Proposed: Give the fraction in each cohort and cite a supplementary figure.

> **Reviewer note**
>
> Problem: The two analyses are described but no result is shown. Proposed: Cite a supplementary figure or table showing the signatures after filtering, or write "we did not observe changes in the dominant signatures".

> **Comment 5, Mo Liu, 2026-09-06**
>
> New paragraph added by Mo to discuss the potential caller batch effects

Importantly, the 89-type system is not uniformly superior in all settings. Although it improves resolution for indels occurring in long poly-T tracts, it also compresses indels of length greater than or equal to 2 bp, and in some cases may merge biologically distinct patterns that are separated in the 83-type framework. For example, the relationships between InsDel_C and InsDel13, as well as between InsDel11 and InsDel16, suggest that some signatures with distinct sequence preferences in the 83-type system are represented together in the 89-type taxonomy. Taken together, these results suggest that although the 89-type taxonomy represents an important advance, further refinement of indel classification will be needed to fully capture the diversity of indel mutational processes. The unified framework presented here should therefore support more consistent interpretation of current signatures and provide a foundation for future development of an even more informative indel taxonomy.

Materials and methods

Data sources

Somatic indel calls from the ICGC/TCGA (International Cancer Genome Consortium/The Cancer Genome Atlas) Pan-Cancer Analysis of Whole Genomes Consortium and associated clinical characteristics were obtained from the ICGC data portal (https://dcc.icgc.org/releases/current/Projects/, which has been retired. The data were downloaded on 9 May, 2024). Hartwig Medical Foundation (Hartwig) data can be requested at at https://www.hartwigmedicalfoundation.nl/en/appyling-for-data/. Information such as cancer type, age, and sex assoicated with the Hartwig data were taken from Table S16 in Priestley et al. (2019) and from the supplementary tables of Martínez-Jiménez et al. (2023). Table S17 in this study present the aggregated metadata used herein.

De novo Mutational signature extraction

De novo mutational signature extraction was performed using two complementary tools applied to the 83-type, 89-type, and 476-type mutational spectra. A stratified strategy was adopted to capture rare and cancer-specific signatures, encompassing analyses of all genomes, specific cancer types, high-TMB samples, and the PCAWG and Hartwig cohorts individually.

For the hierarchical Dirichlet process–based approach, mSigHdp (v2.1.2) was used. When analyzing all samples, cancer type was used to define the hierarchical structure of the model. For analyses restricted to individual cancer types or to genomes with high tumor mutational burden, de novo

signatures were extracted using two-layer hierarchical Dirichlet process (HDP) mixture models. In all mSigHdp analyses, the following parameters were applied: seedNumber = 1234, burnin = 1000, burnin.multiplier = 20, post.n = 200, post.space = 100, num.child.process = 20, gamma.alpha = 1, and gamma.beta = 50. For analyses including all samples or genomes with high tumor mutational burden, an additional downsampling step was applied (downsample_threshold = 3000) to reduce computational time.

In parallel, de novo mutational signatures were extracted using the Non-negative matrix factorization based approach (NMF), SigProfilerExtractor (v1.1.24). For analyses including all genomes, NMF was performed with the number of signatures evaluated between 10 and 40 (minimum_signatures = 10; maximum_signatures = 40). For analyses restricted to individual cancer types or to genomes with high tumor mutational burden, the number of signatures evaluated ranged from 2 to 20 (minimum_signatures = 2; maximum_signatures = 20). Each factorization was repeated 100 times (nmf_replicates = 100), with all other parameters set to their recommended default values. Most signatures were extracted at the "Optimal Solution" identified by SigProfilerExtractor, although several rare signatures were identified in alternative solutions.

Match 83-type and 89-type signatures into reference signatures

For 83-type signatures, the signatures were matched to previously identified COSMIC signatures (v3.6, https://cancer.sanger.ac.uk/signatures/id/). For 89-type signatures, the signatures supported by both mSigHdp and SigProfilerExtractor were matched to the signatures reported in Koh et al. A detailed comparison can be found in Table 1.

We compared all *de novo* signatures supported by both mSigHdp and SigProfilerExtractor to the published signatures and categorized them into three groups: (1) known signature: if an extracted signature has a cosine similarity of ≥ 0.85 with a COSMIC signature; (2) merged signatures: if an extracted signatures can be reconstructed by at most 3 COSMIC signatures with a reconstructed similarity of ≥ 0.9; (3) novel signatures: the extracted signatures do not fit into the known signatures or the merged signatures. We focused on (1) and (3) in this study.

Assignment of signature activity to each individual genome

To leverage the resolution of both the 83-type and 89-type mutational signatures, we constructed a unified 172-type reference set and applied SigProfilerAssignment to decompose the 172-type spectra in low-TMB tumors (indels $< 5{,}000$). In contrast, high-TMB tumors (indels $\geq 5{,}000$), as well as tumors where 1-bp T insertions or deletions in at least five poly-T tracts account for $\geq$50% of the mutations (Table S17), often exhibited highly sparse spectra. This sparsity further masks other mutational signals. Therefore regarding these cases, we employed mSigAct, estimating likelihoods under a multinomial distribution. This was performed in a two-step computational framework. First, we applied the find_best_reconstruction_QP function from the mSigTools R package (v1.0.7), which performs fast signature attribution via quadratic programming optimization. Second, we refined these initial estimates using the PresenceAttributeSigActivity function from the mSigAct R package. Final signature assignments were obtained by integrating results from SigProfilerAssignment for low-TMB tumors and mSigAct for high-TMB tumors.

Note on Table S11 totals: The column sums of Table S11 are not on one scale. For 6,472 samples, the sum is exactly twice the number of indels in the sample's 89-type catalog (Example_data/sigprofiler.input.89type.catalog/example.89type.All.txt in the code repo). These are the samples fit with SigProfilerAssignment to the 172-type catalog, which stacks the 83-type and 89-type catalogs and so counts each indel twice. For the other 503 samples (195 with at least 5,000 indels and 308 with fewer, presumably the poly-T-dominated tumors), the sum equals the indel count. These are presumably the samples fit with mSigAct. Example: CPCT02340014T has 3,260 indels but a Table S11 sum of 6,520. Consequences: (1) Table S11 exposures cannot be compared across the two groups without halving the SigProfilerAssignment ones. (2) The Figure 5 panel C rule "total indels > 5,000" (explore-attributions/reproduce_fig5_panelC_with_MSS_hyper.R, which uses Table S11 column sums) is in effect "> 2,500 indels" for most samples. Using the catalog totals, 195 samples have more than 5,000 indels, while 566 samples have a Table S11 sum above 5,000. (3) The project CLAUDE.md statement that Table S11 column sums equal total indels per sample is wrong for most samples. Not yet confirmed which samples went through mSigAct.

Simulating synthetic cancer datasets

We generated simulated indels for each tumor with SigProfilerSimulator version 1.2.2 (Bergstrom et al. 2020) in the 83-type classification, with 10 simulations per tumor (69,750 simulated genomes in total), on genome build [BUILD] and with [SETTINGS, for example which features of each tumor's mutations were preserved]. Each simulated indel was given the signature probabilities of the real indels of the same tumor and 83-type indel class. Simulated indels in a class that did not occur among the tumor's real indels, and simulated indels from tumors without signature probabilities, were removed. The simulated indels were then annotated for transcription strand, replication strand, and genic versus intergenic region in the same way as the real indels. The simulated genomes (about 3 GB compressed) are available on request.

Annotating somatic indels based on transcribed versus untranscribed strand

We followed the method in Otlu et al. (2023). Briefly, somatic indels were called with respect to the + strand of the reference genome and further annotated in regard to the pyrimidine base(s) of the insertion/deletion. Thus, indels with only C or T bases were annotated as + strand mutations; indels with only A or G bases were annotated as – strand mutations. The remaining indels were not included in the analysis. For example, deletion of a T from a poly-T/poly-A tract was assigned to the strand carrying the T's, and deletion of a C from a poly-C/poly-G tract was assigned to the strand carrying the C's. Next, + strand indels in protein coding genes were further subclassified as untranscribed (sense) if the gene was on the + strand of the genome, or else transcribed (template). The logic was inverted for – strand indels. Indels in bidirectionally transcribed regions were ignored.

Note: The previous text said "+ strand indels ... were further subclassified as transcribed (template) if the gene's sense strand was on the + strand of the genome, or else untranscribed (sense)". That is the reverse of the code. In ID83_topography_analysis_code, r.03_mergeInf.R computes new.trans.strand = trans.strand × Strand, where trans.strand is the gene's strand and Strand is the pyrimidine strand (+, -, or Q for mixed), and r.06_compteOddsRatio.R labels +1 as UnTranscribed

and -1 as Transcribed. So a C/T indel on the + strand in a + strand gene is on the sense (untranscribed) strand, which is the standard convention. The code that computes the Strand column (/public/data/Topography_analysis/reference/single.indel.aggregate.strand.added.matrix.txt) is not in the code repo, so the "only C or T bases" rule above cannot be checked there. Confirm it with Mo, including which bases are examined (the inserted or deleted sequence only, or also the repeat unit).

Annotating somatic indels as genic or intergenic

Indels were annotated with the R package ICAMS (https://github.com/steverozen/ICAMS). An indel was labeled genic if it fell within the transcribed range (from the start to the end of the gene, including introns) of a gene with a Consensus CDS (CCDS) ID in GENCODE release 30 [(lifted over to GRCh37, if that is the genome build used)], on either strand. All other indels were labeled intergenic. The same gene annotation provides the gene strand used for the transcription-strand analysis.

Annotating somatic indels based on leading versus lagging replication strand

Replication strand was determined by wavelet-smoothed replication-timing signal data that indicated both "valleys" (replication termination zones) and "peaks" (replication initiation zones) (https://hgdownload.cse.ucsc.edu/goldenPath/hg19/encodeDCC/wgEncodeUwRepliSeq/). Valleys and peaks were sorted by the genomic coordinate in ascending order. In regard to + strand of the reference genome, replication timing signals were examined for consecutive stretches of the genome (from valley to peak or from peak to valley), with positive slope corresponded to leading strand regions and negative slope corresponded to lagging strand regions. Then for the - strand of the reference genome, leading regions (- slope) and lagging regions (+slope) were automatically acquired. Similar to the annotation for transcription, indels were first annotated as + or – strand mutations based on the pyrimidine bases. Next, indels were counted as being on leading strand or lagging strand based on their occupancy in a leading or lagging region.

Detecting strand asymmetries across cancer types

Strand asymmetry analyses were based on the assignment of signature probabilities to each individual indel mutation. Only indels with probability greater than 0.5 of being due to a given indel signature were retained. For each indel signature and for all cancer types having this mutational signature, we retrieved the number of indels on each strand/region. For each feature, a cancer type was evaluated for a signature only if it had at least 1,000 real indels attributed to that signature (probability > 0.5) that could be assigned to one of the two categories of the feature. Figure 5 and Figure S3 show only evaluated cancer types. Table S12 reports the Fisher's exact test results for all cancer types and signatures.

For each strand asymmetry analyses (genic and intergenic region asymmetry, transcription strand asymmetry, replication strand asymmetry), indel mutations were split into two types and counted (genic vs intergenic mutation, leading strand vs lagging strand mutation, untranscribed strand vs transcribed strand mutation). The two types were denoted as +/- strand mutations in all three

cases. The ratio of real somatic indels and the ratio of simulated somatic indels was calculated separately:

strand ratio = (+ strand_mutation_counts)·(- strand_mutation_counts)$^{-1}$

We calculated the odds ratio between the strand ratio of real somatic indels and the strand ratio of simulated somatic indels.

We tested each cancer type and signature for asymmetry with Fisher's exact test on the 2×2 table of real and simulated indel counts (Table S12 reports the P values and the conditional maximum-likelihood odds ratios from this test). For Figure 5 and Figure S3, the magnitude of asymmetry is the ratio of the real strand ratio to the simulated strand ratio, inverted when needed so that it is ≥ 1, together with the direction of the asymmetry. A cancer type was considered to show asymmetry for a signature if $P < 0.05$. Figure 5 summarizes each signature over the cancer types with $P < 0.05$: it is labeled with the shared direction if they all agree, as "Mix" if they do not, and is not shown if no cancer type has $P < 0.05$ and an odds ratio ≥ 1.1. P values were not adjusted for multiple testing.

For example, for genic versus intergenic regions in one cancer type, the 2×2 table holds the numbers of real and of simulated indels attributed to the signature that fall in genic and in intergenic regions.

Analyses of replication timing across cancer types

Replication timing data were obtained from (https://genomebiology.biomedcentral.com/articles/10.1186/s13059-018-1509-y) The replication time signals were sorted in a descending order and subsequently divided into deciles. Somatic indels were counted within the corresponding deciles based on their overlap with the replication domains in the examined deciles. As in the strand asymmetry analyses, only indels with probability greater than 0.5 of being due to a given signature were counted. For each signature, only cancer types with at least 500 such real indels were analyzed. Counts in the other cancer types were set to 0.

For each signature and cancer type, three linear regression models were fitted across the 10 deciles (lm models 1 to 3 in Table S13), where $R = (R_1, ..., R_{10})$ are the real indel counts, $S = (S_1, ..., S_{10})$ are the simulated indel counts, and $c = (1, 2, ..., 10)$ is the decile: $lm(R \sim c)$, $lm(S \sim c)$, and $lm(R \sim c + S)$. A slope or coefficient was considered significant if $P < 0.05$. The trend of real indels relative to simulated indels was then classified as follows. If only the real slope was significant, the trend was increasing or decreasing according to its sign. If only the simulated slope was significant, the trend was the opposite of the simulated slope's sign. If both were significant with opposite signs, the trend followed the real slope. If both were significant with the same sign, the trend followed the sign of the decile coefficient in the third model, or was flat if that coefficient was not significant. If neither slope was significant, or the cancer type had fewer than 500 real indels for the signature, no trend was assigned ("None" in Table S13). "Increasing" means increasing from early to late replicating regions.

MSI/MSS status and high/low TMB status

NEEDS DRASTIC CUTTING [For PCAWG genomes, the MSI (microsatellite instability) status was evaluated by the PCAWG working group and obtained from the synapse repository (https:

//www.synapse.org/#!Synapse:syn8016399, the data was downloaded on May 2022). For Hartwig genomes, the MSI status was downloaded from the supplementary data of Priestley et al., 2019. The genomes with >4,000 IDs were labeled as high TMB tumors. The thresholds were selected based on the minimum number of mutations of the pre-defined MSI tumors. We then used MSIseq to predict the MSI status of high TMB tumors.[{.mark}]

Extended sequence context analysis

To analyze a specific signature and indel type of interest, we first identified the 5 genomes with the highest contribution of the corresponding signature activity. From these genomes, we extracted all indels of the relevant type. We then examined the nucleotide sequence within a 21-base pair window centered on each indel site ($\pm$10 nucleotides from the indel position). For each position within this window, we calculated the frequency of each nucleotide (A, T, C, and G). The logo was plotted based on the frequency matrix by seqLogo function of seqLogo R package (version 1.71.0)

Funding

The work was supported by the Young Scientists Fund of the National Natural Science Foundation of China (Grant No. 32500546), Natural Science Foundation of Guangdong Province, China (Grant No. 2026A1515010445), International Science and Technology Innovation Cooperation between Governments for National Key R&D Program Projects (2023YFE0107700), Guangzhou Municipal Science and Technology Bureau (2025A04J4304), Shenzhen Science and Technology Innovation Program (JCYJ20250604174809012).

Author contributions

M.L. and S.G.R. conceived the project and designed the analyses. M.L. and S.G.R. designed computational analysis. M.L., Q.Z. and Y.Y. performed mutational signature extraction, attribution and characterization computational analysis. M.H., and R.Y. characterized the topography of 83-type mutational signatures. M.L., X.M.W., R.S. and S.G.R. designed the website. Data interpretation and manuscript write-up were carried out by M.L., M.H., and S.G.R., with contributions from all other authors. All authors reviewed and approved the final manuscript.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Claude in order to polish the text and ensure grammatical accuracy and appropriate vocabulary usage. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

References

Alexandrov, Ludmil B., Young Seok Ju, Kerstin Haase, Peter Van Loo, Iñigo Martincorena, Serena Nik-Zainal, Yasushi Totoki, et al. 2016. "Mutational Signatures Associated with Tobacco Smoking in Human Cancer." *Science* 354 (6312): 618–22. https://doi.org/10.1126/science.aag0299.

Alexandrov, Ludmil B., Jaegil Kim, Nicholas J. Haradhvala, Mi Ni Huang, Alvin Wei Tian Ng, Yang Wu, Arnoud Boot, et al. 2020. "The Repertoire of Mutational Signatures in Human Cancer." *Nature* 578 (7793): 94–101. https://doi.org/10.1038/s41586-020-1943-3.

Alexandrov, Ludmil B., Serena Nik-Zainal, David C. Wedge, and Samuel A. J. R. Aparicio. 2014. "Signatures of Mutational Processes in Human Cancer" 500 (7463): 415–21. https://doi.org/10.1038/nature12477.Signatures.

Bergstrom, Erik N., Mark Barnes, Iñigo Martincorena, and Ludmil B. Alexandrov. 2020. "Generating Realistic Null Hypothesis of Cancer Mutational Landscapes Using SigProfilerSimulator." *BMC Bioinformatics* 21 (1). https://doi.org/10.1186/s12859-020-03772-3.

Boot, Arnoud, Mi Ni Huang, Alvin W. T. Ng, Szu Chi Ho, Jing Quan Lim, Yoshiiku Kawakami, Kazuaki Chayama, Bin Tean Teh, Hidewaki Nakagawa, and Steven G. Rozen. 2018. "In-Depth Characterization of the Cisplatin Mutational Signature in Human Cell Lines and in Esophageal and Liver Tumors." *Genome Research* 28 (5): 654–65. https://doi.org/10.1101/gr.230219.117.

Boot, Arnoud, Mo Liu, Nicole Stantial, Viraj Shah, Willie Yu, Karin C. Nitiss, John L. Nitiss, Sue Jinks-Robertson, and Steven G. Rozen. 2022. "Recurrent Mutations in Topoisomerase IIα Cause a Previously Undescribed Mutator Phenotype in Human Cancers." *Proceedings of the National Academy of Sciences* 119 (4): e2114024119. https://doi.org/10.1073/pnas.2114024119.

Boot, Arnoud, Alvin W. T. Ng, Fui Teen Chong, Szu Chi Ho, Willie Yu, Daniel S. W. Tan, N. Gopalakrishna Iyer, and Steven G. Roze. 2020. "Characterization of Colibactin-Associated Mutational Signature in an Asian Oral Squamous Cell Carcinoma and in Other Mucosal Tumor Types." *Genome Research* 30 (6): 803–13. https://doi.org/10.1101/gr.255620.119.

Caipa Garcia, Angela L., Jill E. Kucab, Halh Al-Serori, Rebekah S. S. Beck, Madjda Bellamri, Robert J. Turesky, John D. Groopman, et al. 2024. "Tissue Organoid Cultures Metabolize Dietary Carcinogens Proficiently and Are Effective Models for DNA Adduct Formation." *Chemical Research in Toxicology* 37 (2): 234–47. https://doi.org/10.1021/acs.chemrestox.3c00255.

Chen, Lei, Chong Zhang, Ruidong Xue, Mo Liu, Jian Bai, Jinxia Bao, Yin Wang, et al. 2024. "Deep Whole-Genome Analysis of 494 Hepatocellular Carcinomas." *Nature*, March. https://doi.org/10.1038/s41586-024-07054-3.

Cooper, David N, Matthew Mort, Peter D Stenson, Edward V Ball, and Nadia A Chuzhanova. 2010. "Methylation-Mediated Deamination of 5-Methylcytosine Appears to Give Rise to Mutations Causing Human Inherited Disease in CpNpG Trinucleotides, as Well as in CpG Dinucleotides."

Davies, Helen, Dominik Glodzik, Sandro Morganella, Lucy R. Yates, Johan Staaf, Xueqing Zou, Manasa Ramakrishna, et al. 2017. "HRDetect Is a Predictor of BRCA1 and BRCA2 Deficiency Based on Mutational Signatures." *Nature Medicine* 23 (4): 517–25. https://doi.org/10.1038/nm.4292.

Degasperi, Andrea, Xueqing Zou, Tauanne Dias Amarante, Andrea Martinez-Martinez, Gene Ching Chiek Koh, João M. L. Dias, Laura Heskin, et al. 2022. "Substitution Mutational Signatures in Whole-Genome–Sequenced Cancers in the UK Population." *Science* 376 (6591). https://doi.org/10.1126/science.abl9283.

DeWeerd, Rachel A., Eszter Németh, Ádám Póti, Nataliya Petryk, Chun-Long Chen, Olivier Hyrien, Dávid Szüts, and Abby M. Green. 2022. "Prospectively Defined Patterns of APOBEC3A Mutagenesis Are Prevalent in Human Cancers." *Cell Reports* 38 (12): 110555. https://doi.org/10.101

6/j.celrep.2022.110555.
Dziubańska-Kusibab, Paulina J., Hilmar Berger, Federica Battistini, Britta A. M. Bouwman, Amina Iftekhar, Riku Katainen, Tatiana Cajuso, et al. 2020. "Colibactin DNA-damage Signature Indicates Mutational Impact in Colorectal Cancer." *Nature Medicine* 26 (7): 1063–69. https://doi.org/10.1038/s41591-020-0908-2.
Grolleman, Judith E., Richarda M. de Voer, Fadwa A. Elsayed, Maartje Nielsen, Robbert D. A. Weren, Claire Palles, Marjolijn J. L. Ligtenberg, et al. 2019. "Mutational Signature Analysis Reveals NTHL1 Deficiency to Cause a Multi-tumor Phenotype." *Cancer Cell* 35 (2): 256–266.e5. https://doi.org/10.1016/j.ccell.2018.12.011.
Hoang, Margaret L., Chung-Hsin Chen, Viktoriya S. Sidorenko, Jian He, Kathleen G. Dickman, Byeong Hwa Yun, Masaaki Moriya, et al. 2013. "Mutational Signature of Aristolochic Acid Exposure as Revealed by Whole-Exome Sequencing." *Science Translational Medicine* 5 (197). https://doi.org/10.1126/scitranslmed.3006200.
Huang, Mi Ni, John R. McPherson, Ioana Cutcutache, Bin Tean Teh, Patrick Tan, and Steven G. Rozen. 2015. "MSIseq: Software for Assessing Microsatellite Instability from Catalogs of Somatic Mutations." *Scientific Reports* 5 (1): 13321. https://doi.org/10.1038/srep13321.
Huang, Mi Ni, Willie Yu, Wei Wei Teoh, Maude Ardin, Apinya Jusakul, Alvin Wei Tian Ng, Arnoud Boot, et al. 2017. "Genome-Scale Mutational Signatures of Aflatoxin in Cells, Mice, and Human Tumors." *Genome Research* 27 (9): 1475–86. https://doi.org/10.1101/gr.220038.116.
Islam, S. M.Ashiqul, Marcos Díaz-Gay, Yang Wu, Mark Barnes, Raviteja Vangara, Erik N. Bergstrom, Yudou He, et al. 2022. "Uncovering Novel Mutational Signatures by de Novo Extraction with SigProfilerExtractor." *Cell Genomics* 2 (11). https://doi.org/10.1016/j.xgen.2022.100179.
Jin, Hu, Doga C. Gulhan, Benedikt Geiger, Daniel Ben-Isvy, David Geng, Viktor Ljungström, and Peter J. Park. 2024. "Accurate and Sensitive Mutational Signature Analysis with MuSiCal." *Nature Genetics* 56 (3): 541–52. https://doi.org/10.1038/s41588-024-01659-0.
Koh, Gene Ching Chiek, Arjun Scott Nanda, Giuseppe Rinaldi, Soraya Boushaki, Andrea Degasperi, Cherif Badja, Andrew Marcel Pregnall, et al. 2025. "A Redefined InDel Taxonomy Provides Insights into Mutational Signatures." *Nature Genetics*, April. https://doi.org/10.1038/s41588-025-02152-y.
Kucab, Jill E., Xueqing Zou, Sandro Morganella, Madeleine Joel, A. Scott Nanda, Eszter Nagy, Celine Gomez, et al. 2019. "A Compendium of Mutational Signatures of Environmental Agents." *Cell* 177 (4): 821–836.e16. https://doi.org/10.1016/j.cell.2019.03.001.
Lawson, Andrew R. J., Federico Abascal, Tim H. H. Coorens, Yvette Hooks, Laura O'Neill, Calli Latimer, Keiran Raine, et al. 2020. "Extensive Heterogeneity in Somatic Mutation and Selection in the Human Bladder." *Science* 370 (6512): 75–82. https://doi.org/10.1126/science.aba8347.
Liu, Mo, Yang Wu, Nanhai Jiang, Arnoud Boot, and Steven G Rozen. 2023. "mSigHdp: Hierarchical Dirichlet Process Mixture Modeling for Mutational Signature Discovery." *NAR Genomics and Bioinformatics* 5 (1): lqad005. https://doi.org/10.1093/nargab/lqad005.
Martínez-Jiménez, Francisco, Ali Movasati, Sascha Remy Brunner, Luan Nguyen, Peter Priestley, Edwin Cuppen, and Arne Van Hoeck. 2023. "Pan-Cancer Whole-Genome Comparison of Primary and Metastatic Solid Tumours." *Nature* 618 (7964): 333–41. https://doi.org/10.1038/s41586-023-06054-z.
Mellon, Isabel, Graciela Spivak, and Philip C. Hanawalt. 1987. "Selective Removal of Transcription-Blocking DNA Damage from the Transcribed Strand of the Mammalian DHFR Gene." *Cell* 51 (2): 241–49. https://doi.org/10.1016/0092-8674(87)90151-6.
Moody, Sarah, Sergey Senkin, S. M. Ashiqul Islam, Jingwei Wang, Dariush Nasrollahzadeh, Ricardo Cortez Cardoso Penha, Stephen Fitzgerald, et al. 2021. "Mutational Signatures in Esophageal Squamous Cell Carcinoma from Eight Countries with Varying Incidence." *Nature Genetics* 53:

1553–63. https://doi.org/10.1038/s41588-021-00928-6.
Ng, Alvin W T, Song Ling Poon, Mi Ni Huang, Jing Quan Lim, Arnoud Boot, Willie Yu, Yuka Suzuki, et al. 2017. "Aristolochic Acids and Their Derivatives Are Widely Implicated in Liver Cancers in Taiwan and Throughout Asia." https://doi.org/10.1126/scitranslmed.aan6446.
Nik-Zainal, Serena, Ludmil B. Alexandrov, David C. Wedge, Peter Van Loo, Christopher D. Greenman, Keiran Raine, David Jones, et al. 2012. "Mutational Processes Molding the Genomes of 21 Breast Cancers." *Cell* 149 (5): 979–93. https://doi.org/10.1016/j.cell.2012.04.024.
Otlu, Burçak, Marcos Díaz-Gay, Ian Vermes, Erik N. Bergstrom, Maria Zhivagui, Mark Barnes, and Ludmil B. Alexandrov. 2023. "Topography of Mutational Signatures in Human Cancer." *Cell Reports* 42 (8): 112930. https://doi.org/10.1016/j.celrep.2023.112930.
Poon, Song Ling, Mi Ni Huang, Yang Choo, John R McPherson, Willie Yu, Hong Lee Heng, Anna Gan, et al. 2015. "Mutation Signatures Implicate Aristolochic Acid in Bladder Cancer Development." *Genome Medicine* 7 (1): 38. https://doi.org/10.1186/s13073-015-0161-3.
Poon, Song Ling, See-Tong Pang, John R. McPherson, Willie Yu, Kie Kyon Huang, Peiyong Guan, Wen-Hui Weng, et al. 2013. "Genome-Wide Mutational Signatures of Aristolochic Acid and Its Application as a Screening Tool." *Science Translational Medicine* 5 (197). https://doi.org/10.1126/scitranslmed.3006086.
Priestley, Peter, Jonathan Baber, Martijn P. Lolkema, Neeltje Steeghs, Ewart de Bruijn, Charles Shale, Korneel Duyvesteyn, et al. 2019. "Pan-Cancer Whole-Genome Analyses of Metastatic Solid Tumours." *Nature* 575 (7781): 210–16. https://doi.org/10.1038/s41586-019-1689-y.
Reijns, Martin A. M., David A. Parry, Thomas C. Williams, Ferran Nadeu, Rebecca L. Hindshaw, Diana O. Rios Szwed, Michael D. Nicholson, et al. 2022. "Signatures of TOP1 Transcription-Associated Mutagenesis in Cancer and Germline." *Nature* 602 (7898): 623–31. https://doi.org/10.1038/s41586-022-04403-y.
Riva, Laura, Arun R. Pandiri, Yun Rose Li, Alastair Droop, James Hewinson, Michael A. Quail, Vivek Iyer, et al. 2020. "The Mutational Signature Profile of Known and Suspected Human Carcinogens in Mice." *Nature Genetics* 52 (11): 1189–97. https://doi.org/10.1038/s41588-020-0692-4.
Vrieling, H., M. L. Van Rooijen, N. A. Groen, M. Z. Zdzienicka, J. W. Simons, P. H. Lohman, and A. A. van Zeeland. 1989. "DNA Strand Specificity for UV-Induced Mutations in Mammalian Cells." *Molecular and Cellular Biology* 9 (3): 1277–83. https://doi.org/10.1128/mcb.9.3.1277.

Table 1

Table 1: Summary of the correspondences and cosine similarities between the 89-type indel signatures (i) signatures from the 476-type collapsed set, (ii) the tumor spectra with highest cosine similarities, (iii) the 89-type signatures reported in Koh et al., (iv) and 83-type signatures extracted in this study along with 83-type signatures reported in Jin et al.

89-Type Signature ID	Cosine 476-Type Sig Collapsed to 89-Type	Cosine 89-Type Sig vs Best Match Spectrum	Koh et al. Sig	Cosine 89-Type Sig vs Koh et al.	83-Type Sig ID	Jin et al. Sig	Cosine vs Jin et al.
InsDel1a	0.9923	0.9831	InD14	0.9023	C_ID1	ID1*	0.9849
InsDel1b	0.9925	0.9850	InD15	0.9796	C_ID1	ID1	0.9849
InsDel1c	0.9819	0.9675	InD14*	0.9415	C_ID1	ID1	0.9849
InsDel2a	0.9853	0.9710	InD2a	0.9824	C_ID2	ID2*	0.9932
InsDel2b	0.9858	0.9645			C_ID2	ID2	0.9932
InsDel2c	0.9842	0.9744			C_ID2	ID2	0.9932
InsDel3a	0.9515	0.9738			C_ID3	ID3*	0.9403
InsDel3b	0.9970	0.9952	InD3b	0.9527	C_ID3	ID3	0.9403
InsDel4	0.9955	0.9845	InD4a	0.9820	C_ID4		
InsDel5a	0.9983	0.9908	InD29		C_ID5		
InsDel5b	0.9953	0.9826			C_ID5		
InsDel5c	0.9575	0.9344			C_ID5		
InsDel6	0.9981	0.9949	InD6	0.9745	C_ID6	ID6	0.9156
InsDel7	0.9977	0.9967	InD7	0.9797	C_ID7**		
InsDel8	0.9942	0.9914	InD8	0.9812	C_ID8	ID8	0.9683
InsDel9	0.9992	0.9259	InD9b	0.9170	C_ID9		
InsDel10	0.9569	0.8457			C_ID10**	ID10	0.9016
InsDel11	1.0000	1.0000			C_ID11	ID11a	0.9975
InsDel12	0.9908	0.9701			C_ID12	ID12	0.9368
InsDel13	0.9962	0.9908	InD13*	0.9886	C_ID13	ID13	0.9580
InsDel14	0.9881	0.9945	InD12	0.9831	C_ID14	ID14	0.9584
InsDel15	1.0000	0.9972			C_ID15		
InsDel16	1.0000	1.0000			C_ID16		
InsDel17	1.0000	0.9999			C_ID17		
InsDel18	0.9994	0.9958	InD18	0.9959	C_ID18	ID18	0.9087
InsDel19a	0.9913	0.9494			C_ID19		
InsDel19b	0.9943	0.9922			C_ID19		
InsDel23	0.9945	0.9606			C_ID23		
InsDel_Aα	0.9972	0.8097			ID_A		
InsDel_Aβ	0.9792	0.9361	InD9a	0.9650	ID_A		
InsDel_B	0.9919	0.9872			ID_B		
InsDel_C	0.9968	0.9713	InD13	0.9268	ID_C		
InsDel_D	0.9746	0.9879	InD1	0.9256	ID_D		
InsDel_E	0.9879	0.9776			ID_E		
InsDel_F	0.9993	0.9925			ID_F	ID4	0.9726
InsDel_G	0.9966	0.9547			ID_G	ID11b	0.9467
InsDel_H	0.9540	0.8206			ID_H		
InsDel_I	0.9991	0.9998			ID_I	ID25	0.9323
InsDel_J	0.9753	0.5628			ID_J**		

(*continued*)

89-Type Signature ID	Cosine 476-Type Sig Collapsed to 89-Type	Cosine 89-Type Sig vs Best Match Spectrum	Koh et al. Sig	Cosine 89-Type Sig vs Koh et al.	83-Type Sig ID	Jin et al. Sig	Cosine vs Jin et al.
InsDel_K	0.9964	0.8572	InD20	0.9488	ID_K	ID26	0.9815
InsDel_L	0.9981	0.9658			ID_L		
InsDel_M	0.9942	0.9999			ID_M		
InsDel_Nα	0.9900	0.9840	InD16b	0.9229	ID_N		
InsDel_Nβ	0.9949	0.9850	InD21	0.9047	ID_N		

* This is the signature with the best match to the given 89-type signature.

** indicates that in the linking tumor's spectrum, insertions of single Ts in long poly-T contexts were set to 0 before calculating cosine similarity to the 83-type signature.

Figures

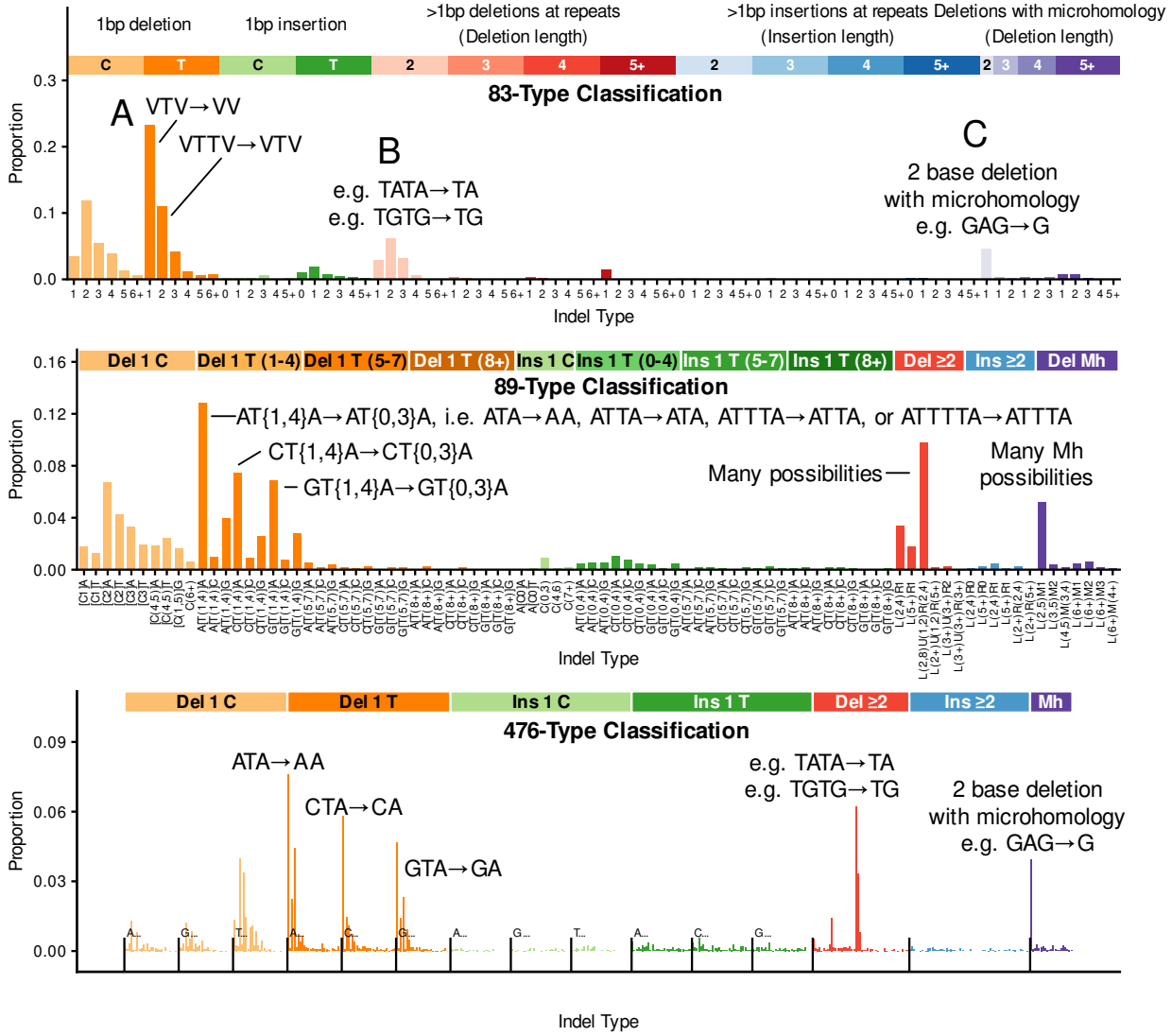


Figure 1: **The aristolochic acid (AA) indel signatures (signature 23) in the three classification schemes.**

(**A**) Looking at labeled bars from left to right, at the 83-type classification highest bar is labeled VTV→VV, indicating a the deletion of a single T (thymine) surrounded by non-thymines (V, the preceding V and following V can be different bases, e.g. ATC→AC). Still in the 83-type classification, the second most frequent deletion is a deletion of single T from TT flanked by non Ts (VTTV→VTV). In the 89-type classification there is a many-to-many representation of these (and other mutations) to a peak labeled AT{1,4}→AT{0,3}A, which indicates 4 possible types of mutations as indicated in the figure ATA→AA, ATTA→ATA, ATTTA→ATTA, and ATTTTA→ATTTA. To the right are analogous deletions from e.g. CTA→CA, CTTA→ CTA, etc, and then further to the right, GTA→GA, GTTA→GTA, etc. Finally, the 476-type classification makes clear that in fact the main deletions are ATA→AA, CTA→CA, and GTA→GA.

(**B**) Further to the left in the 83-type classification there are three light orange bars, with the highest

labeled "e.g. TATA→TA, e.g. TGTG→TG" these bars represent 2-base deletions, with the highest bar representing the deletion of a 2-base unit from the 2-base unit duplicated (e.g. a 2-base unit TA, in the duplicated 2-base unit TATA, with the deletion being TATA→TA). The classification does not capture information on the bases involved. In the 89-type classification, these are captured in the tall red bar labeled "many possibilities", which represents a wide range of deletions, up to deletions such as ATATATAT→AT, which are essentially absent in this signature. The tallest of the corresponding bars in the 476-type classification is labeled "e.g. TATA→TA, e.g. TGTG→TG", in one-to-one correspondence with the light orange bar in the 83-type classification with the same label. Like the 83-type classification, the 476-classification does not capture the identity of the bases involved for these deletions.

(**C**) At the far right, in the 83-type-classification, the light purple bar indicates 2-base deletions with microhomology, e.g. GAG→G, where the G flanking the deletion is the microhomologous base. The bases involved are not captured, and other possibilities include GTG→G, TAT→T, and many others. In the 89-type classification these and many other possible deletions are represented by the peak "Many Mh possibilities", which includes, in addition, deletions such as ATCGTA→A, which are essentially absent from this signature. In the 476-type classification, as in the 83-type classification, the highest bar is labeled "2 base deletions with microhomology", and it represents the same set of mutations as the corresponding bar in the 83-type classification. However, in general, even for the 476 classification there is many-to-many mapping of deletions and insertions of more than 1 base between the 83-type and 476-type classifications (Table S1).

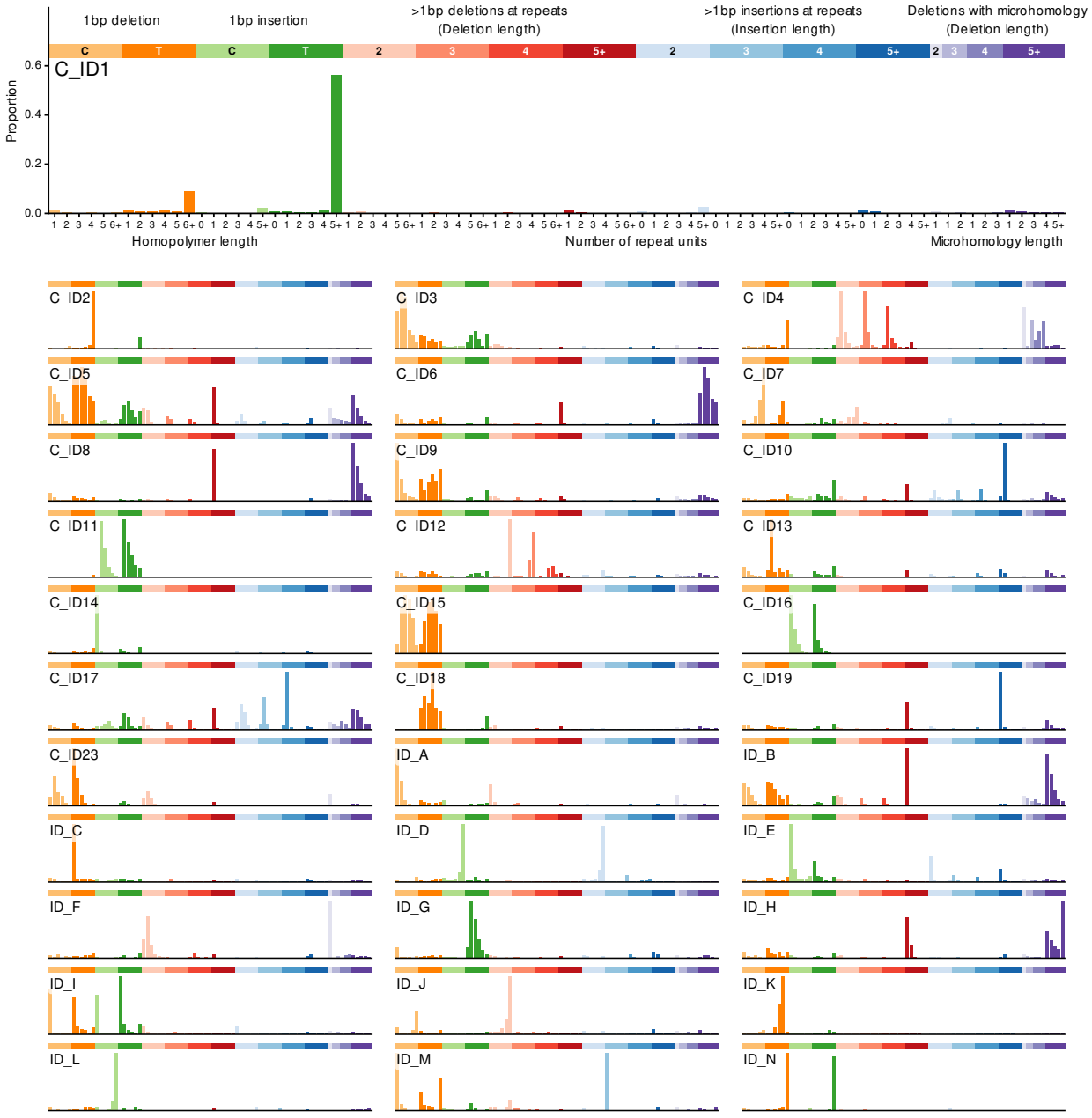


Figure 2: **Profiles of signatures extracted in the 83-type classification scheme**

The x-axes represent indel categories and the y-axes indicate the proportion of each indel category within each signature. C_ID1 is shown with full axis labels, and the remaining signatures are shown as thumbnails on the same scale of categories. Signatures C_ID1 through C_ID23 match reference signatures in COSMIC v3.6 and follow the same numbering. For example, C_ID1 matches ID1. The remaining signatures, ID_A through ID_N, were not present in COSMIC v3.6.



Figure 3: **Profiles of signatures extracted in the 89-type classification scheme**

The x-axes represent indel categories and the y-axes indicate the proportion of each indel category within each signature. InsDel1a is shown with full axis labels, followed by thumbnails of the signatures that correspond to COSMIC reference signatures and then the signatures that are new. For signatures that correspond to COSMIC reference signatures, we use the same numbering and add suffixes (a, b, c) when a single COSMIC signature maps to multiple 89-type signatures. New signatures are lettered (InsDel_A to InsDel_N), with α and β distinguishing two signatures that share a letter. The prefix "InsDel" distinguishes these signatures from those in COSMIC (prefix "ID") and in Koh et al. (2025) (prefix "InD").

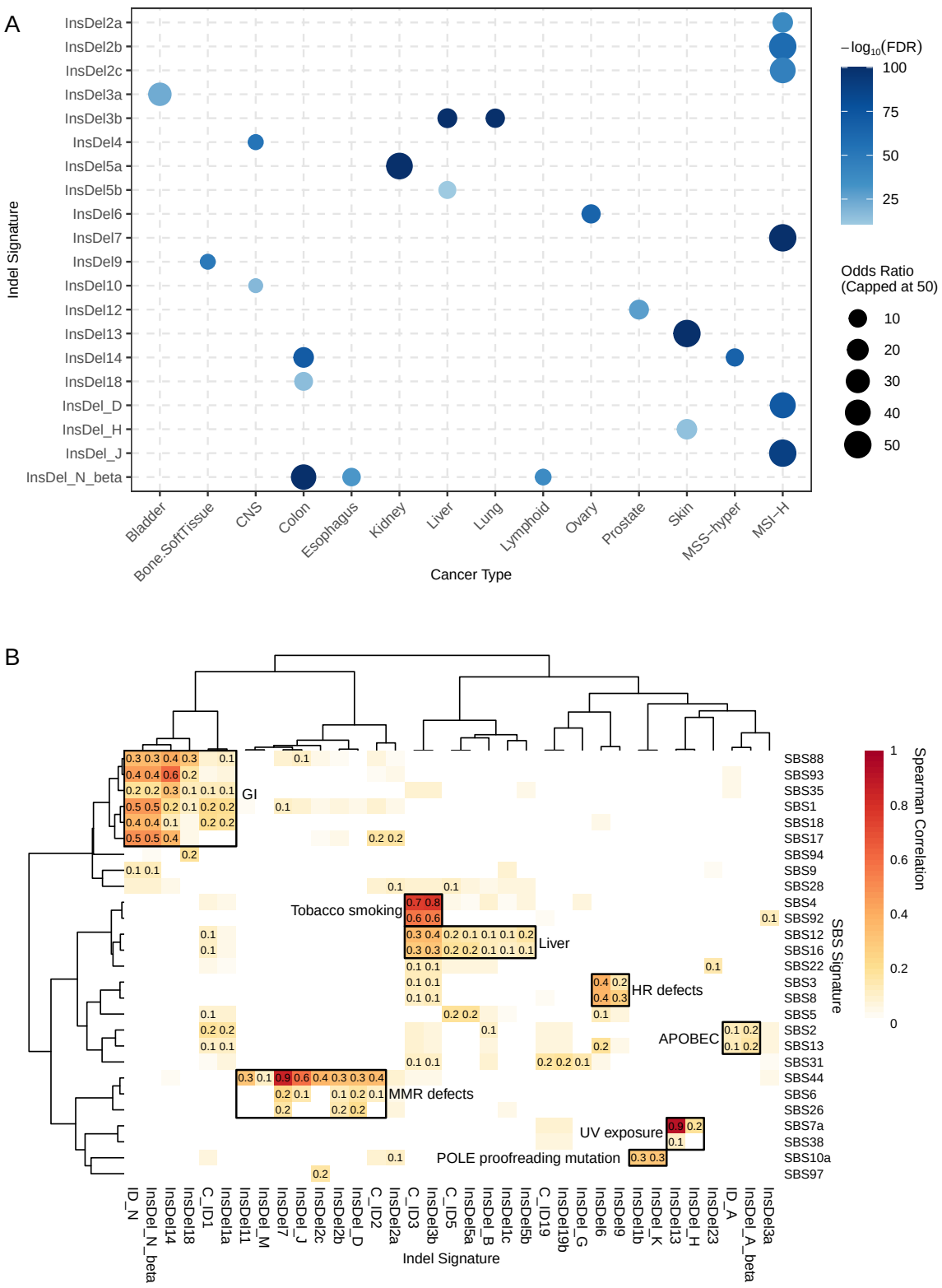


Figure 4: **Association of indel signatures with cancer types and SBS signatures and cancer types**

(**A**) Enrichment for presence of 89-type signatures across cancer types, plus MSI-high MSI-H) and hypermutated (> 4,000 indels) microsatellite stable (MSS) tumors. Only cancer types with odds

ratios > 6 are shown. FDR is the Benjamini-Hochberg false discovery rate based on p values from Fisher's exact tests of he number of tumors with the signature versus the number without in a given tumor type versus all other tumors.

(**B**) The heatmap of Spearman correlation coefficients between indel signatures and various SBS signatures. Correlations that were not significant (Benjamini-Hochberg FDR ≥ 0.05) were set to 0. Signatures were excluded if none of their correlation coefficients was greater than 0.1. Unsupervised hierarchical clustering (Ward's method) of both rows and columns, using 1 – the Pearson correlation between rows (or columns) of the Spearman correlation matrix as distance, identified 8 clusters, boxed and labeled in the figure. When an 83-type signature was not split into several 89-type signatures (for example C_ID14, which corresponds only to InsDel14), only the 89-type signature is shown. GI: Gastrointestinal.



Figure 5: **83-type mutational signatures asymmetries in strands, replication timing and DNA regions.**

(**A**) Transcription strand asymmetries of 83-type signatures. Each row represents one 83-type signature, where n reflects the number of cancer types in which the signature was found. Signatures consistently enriched on the transcribed and untranscribed strands with p values < 0.05 (Fisher's exact test) are shown in circles with red and dark blue colors, respectively. Grey colors indicate inconsistent enrichments in different cancer types. Columns display the odds ratio between the ratio of real mutations and the ratio of simulated mutations (See Methods). Only odds ratios above

1.10 are shown. Circle sizes reflect the proportion of cancer types exhibiting the asymmetry in a signature.

Note on what "transcribed strand" and "untranscribed strand" mean here. We followed Otlu et al. 2023 (https://doi.org/10.1016/j.celrep.2023.112930), which states that mutations "were oriented based on the reference strand and their pyrimidine context". An indel is therefore assigned to the strand that carries its pyrimidine bases. For example, deletion of a T from a poly-T/poly-A tract counts as being on the strand that carries the T's, and deletion of a C from a poly-C/poly-G tract counts as being on the strand that carries the C's (code: r.03_mergeInf.R and r.06_comupteOddsRatio.R in ID83_topography_analysis_code, where new.trans.strand = gene strand × pyrimidine strand, and -1 is labeled Transcribed). Phrases such as "mutations enriched on the transcribed strand" are shorthand for this convention, not a statement about which strand was damaged. For bulky adducts on purines (e.g., tobacco carcinogens on G), the damaged base is on the opposite strand from the pyrimidine. So "enriched on the transcribed strand" means the damaged purine was on the untranscribed strand, consistent with transcription-coupled repair removing damage from the transcribed strand. For UV, the damaged pyrimidines are the counted bases themselves, so "enriched on the untranscribed strand" means the damage was on the untranscribed strand. Otlu et al. use the same shorthand without spelling this out. The text in Results that discusses Figure 5A (C_ID3, C_ID18, C_ID13) should be revised accordingly, and the legend should say which base defines the strand.

(**B**) Replication strand asymmetries of 83-type signatures. Data are presented in a format similar to the one in panel (A), with green and orange colors indicating replication strand asymmetries on the lagging and leading strands, respectively.

(**C**) Enrichment of mutations in genic and intergenic regions for 83-type signatures. Data are presented in a format similar to the one in panel (A), with yellow and light blue colors indicating consistent enrichment in genic and intergenic regions, respectively.

(**D**) 83-type signatures associated with early and late replication timing. Relative mutation counts per replication-timing decile (y axes) are presented for early (left) to late (right) replication domains. Real mutations are shown as bars and simulated mutations as red dashed lines, each scaled to its own maximum. The first four panels sum the cancer types with a trend for the signature, and the numbers above them are the numbers of cancer types in which it increases, decreases, or is flat. Green indicates a signature that increases from early to late replication in every such cancer type (C_ID3). Yellow indicates one that decreases in every such cancer type (C_ID17). Purple indicates one that is flat in more than half of them (ID_K). Blue indicates inconsistent trends (ID_F). The last two panels show ID_F in two single cancer types with opposite trends, bladder (increasing) and skin (decreasing).

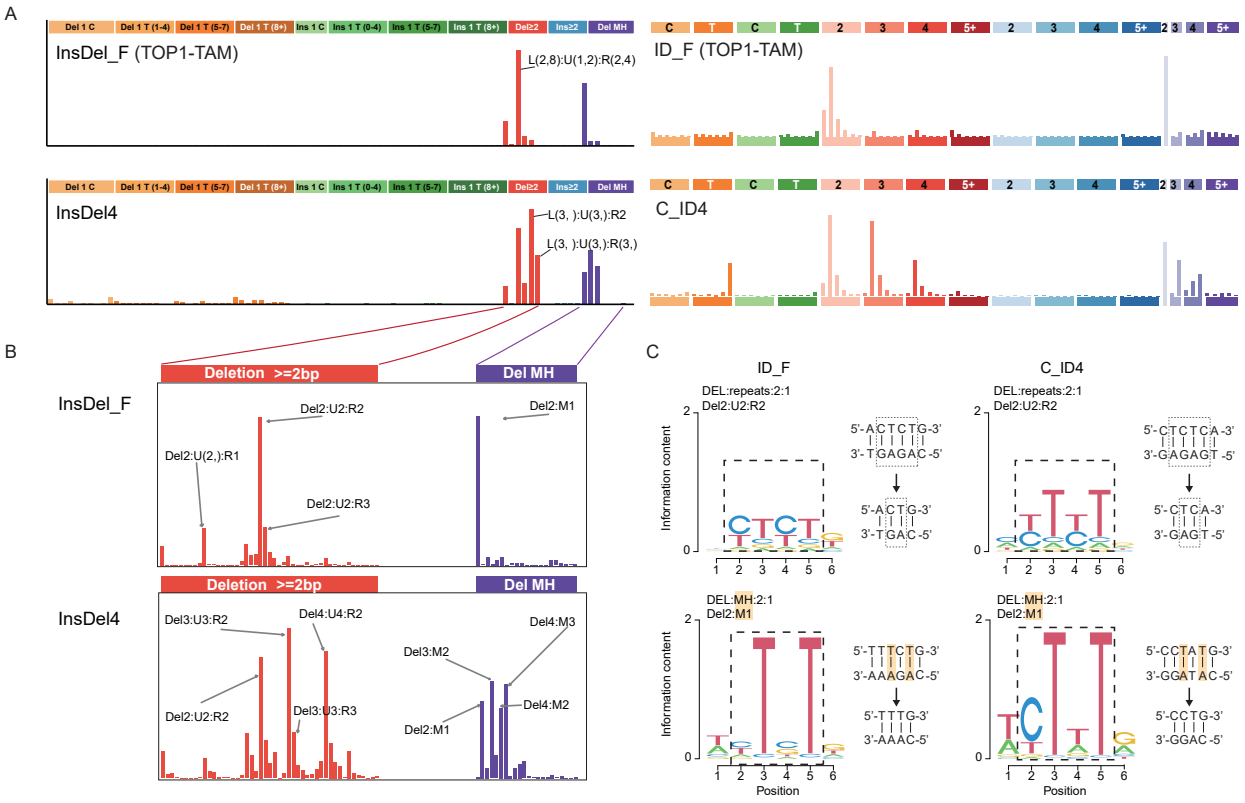


Figure 6: **Novel signature caused by RNase H2 deficiency and topoisomerase-1-associated mutagenesis (TOP1-TAM)**

(**A**) The mutational signature of RNase H2 deficiency (InsDel_F in the 89-type classification, ID_F in the 83-type classification), and a previously identified mutational signature (InsDel4, C_ID4).

(**B**) Deletions of ≥ 2 bases in the 476-type classification scheme in InsDel_F compared to InsDel4. InsDel4 has substantial proportions of deletions of 3 and 4 bases (Del3:U2:R2, Del4:U4:R2, Del3:M2, Del4:M2, Del4:M3) that are essentially absent in InsDel_F. Panel **C** shows Del2:U2:R2 and Del2:M1 examples.

(**C**) Logos of the sequence contexts of 2-base deletions in 5 samples with the highest InsDel_F activity (left) compared to 5 samples with the highest InsDel4 activity (right). The sequence contexts of the 2-base deletions in InsDel_F are on the whole different from the contexts of 2-base deletions in InsDel4. In the lower two sections (DEL:MH:2:1, 2-base deletions with 1 base of microhomology), the microhomologous bases (T:A) are shaded orange.